COLLISION RECORDS, ARITHMETIC, AND RAW LOCAL INVERSION IN A TRIANGULAR LORENTZ GAS

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Abstract. We study the unchanged exact-label raw pointwise local limit and four-coordinate local inversion problem for the triangular Lorentz gas with physical collision time. The existing optical, arithmetic, incidence-load, and exact unvisited-clearance flux arguments are retained. We compute the long-record clearance-to-endpoint Jacobian and prove local positive roof-density bounds which integrate angular folds before taking a roof supremum; a bounded density at each fixed clearance is not required. At a regular critical roof we identify the trajectory as half of an actual reversing periodic orbit. Its exact one-sided density coefficient is $[Rc_* \sqrt{|\det(I - DT_R^{2m})|}]^{-1}$. A quantitative coarea formula keeps arbitrary Borel event cuts and one-sided unvisited boundaries through an explicit angular fraction, without discarding positive boundary contributions. The all-word sum of critical coefficients is bounded relative to the retained geometric-pressure input, but this bound is unscaled. In this revision, exact two-mark coarea on a single physical flight uses the lattice spacing to prove a positive transversality constant $\sqrt{591}/100$; distinct unvisited marks have an exact flux identity and a uniform $O(\delta_1\delta_2)$ factorial bound. The resulting within-flight duplicate-mark mass is $O(\varepsilon^2)$ on the unchanged depth schedule, with all original exact labels. This is a collision-space result and does not imply pointwise roof normalization. Uniform collision-normalized clearance and capacity estimates, the remaining low-incidence-load shortage limit, and the full raw pointwise local limit and inversion conclusion are not proved in this revision.

Introduction

The new two-mark argument is in Section 17. Earlier arguments in Sections 15 and 16 are retained unchanged; they are inherited from the published v114 author revision [1]. They address two interfaces which are not settled by small integrated clearance flux. First, the exact long-record Jacobian in Theorem 15.1 separates a change of coordinates from an unproved long-time density estimate. Theorems 15.4 and 15.5 then give positive local roof estimates, including angular folds with an unbounded fixed-clearance conditional density. All original record tests remain as Borel source selectors.

Second, critical endpoint roofs are not arbitrary exceptional points: Theorems 16.1 and 16.2 identify them with actual reversing periodic orbits and express their density coefficient using the derivative of the genuine physical return. Theorem 16.3 gives the coefficient and a local $O(\sqrt{s})$ error uniformly over Borel selectors, retaining their exact angular fractions. The one-sided action calculation extends to unvisited tangencies and section cuts with positive visited incidences, but no derivative of an undefined singular physical return is asserted. Proposition 16.5 gives a complete finite-count positive sum relative to the expressly retained pressure input. Its lack of an m^{-2} factor is stated, not hidden in a new residual.

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Theorem 17.2 supplies a second-order coarea formula for two distinct unvisited disks on the same genuine free-flight segment. Lemma 17.1 excludes the collision of their angular root Jacobians using only the original triangular-lattice spacing and the original radius interval; Theorem 17.3 gives a uniform quadratic bound. Corollary 17.5 then bounds same-flight duplicate marks for the entire original collision-end depth schedule, under the unchanged exact event, by $8C_{\mathrm{pair}}\varepsilon^2/[c_*(1-r^2)]$ in total variation. This is a new positive physical-source bound, not a claim about m^2-normalized roof essential height. Single unvisited marks, marks on different flights, and capacity deficiency still require independent pointwise estimates.

The preceding clearance-flux extension, preserved here in Sections A–D, supplies exact one-copy mark flux, a grazing-uniform strip geometry, and a stable-test estimate conditional on its growth input. Those are retained results, not new contributions of the present revision. Neither they nor the local coarea results complete the full normalized source comparison. The original physical model, section, radius interval, roof, lattice labels, and ordered limits are unchanged. The proof and source ledgers distinguish complete local proofs, deductions from retained dynamical inputs, and unresolved uniform estimates.

The retained proved results are global roof–angle uniqueness, target-roof marked pressure relative to the explicit geometric inputs, and collision-normalized positive payments for deep, numerous and high-load incidence sources. The complete exact-label raw pointwise local limit assertion (1.5) is still the research target: its equivalent remaining estimate (14.8) is not proved here. The retained load and quadratic-count theorems strengthen the positive source estimates within that same problem, not a replacement problem.

The pointwise raw local limit problem concerns the density at the original target roof, uniformly in collision count, radius and exact return labels. Control of a fixed-window average does not by itself control that density. For the original positive microscopic paired source $\mathsf{U}_B^{\mathrm{micro}}$ and its full nongrazing comparison pool $P_{r_B}^{\mathrm{ng}}$, the estimate left by the preceding argument is

$$\lim_{B\to\infty}\limsup_{m\to\infty}\sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} (\mathsf{U}_{B,m}^{\mathrm{micro}}(t)-K_B P_{r_B,m}^{\mathrm{ng}}(t))_+ = 0. \tag{0.1}$$

This is equivalent, within the retained scalar reduction, to the original pointwise assertion (1.5). It is not yet proved here. The retained source-side contribution is a quantitative payment of a further positive part of this same source and a physical description of every coefficient which survives that payment.

The distinction between source and capacity is essential. On a segment with an actually visited grazing collision at one end, the original paired minimum is zero. Its endpoint maximum, and hence its comparison capacity, need not be zero. The optical and trace arguments retained in Sections 3–6 pay this entire visited-grazing capacity without deleting unvisited tangencies or positive critical and section-boundary traces. They start from the actual broken-path action, use a positive continuant denominator, extend its reduced Hessian continuously to visited grazing, and integrate the inverse gradient along a fixed Gauss angle. All these arguments and their physical hypotheses are retained in the present article.

The source-side argument uses the total variation of the weighted density on an entire physical segment. If c^ε is its endpoint minimum and $\mathcal{V}^\varepsilon$ its variation, we split, before any roof pushforward,

$$c^\varepsilon = \min\{c^\varepsilon, \gamma_B\mathcal{V}^\varepsilon\} + (c^\varepsilon - \gamma_B\mathcal{V}^\varepsilon)_+, \qquad \gamma_B = \max\{2, \varepsilon(B)^{-1/32}\}.$$

The first term can be positive, even when both endpoint traces are positive. Theorem 7.2 pays its complete physical sum by the original uncut weighted volume, with no residence lower bound, word count or comparison reach assumption. The resulting ordered height is $O(B^{-95/384})$.

For the second term, Theorem 8.1 shows that the density stays within relative distance γ_B^{-1} of its endpoint minimum and the original guard stays positive at every point of the closed segment.

In particular, a single point where the original guard vanishes pays the entire original paired coefficient. This assertion is independent of the name of either endpoint mechanism. It does not assert that a single incidence or clearance test witnesses the guard throughout the segment.

Theorem 8.3 retains the full nongrazing comparison pool and gives an exact three-term positive identity for the original microscopic source. The remaining estimate is printed in (8.9), in the original essential-height topology and the original order of limits. Neither nearly constant density on a short segment nor finite-count exhaustion is identified with a proof of this remaining estimate.

Appendix F reconstructs the signed corner cofactor and proves a simultaneous multi-mark expansion. It also states the simultaneous one-sided endpoint lemma separately and makes the uniform constant in the trace modulus explicit. The multi-mark bound is per trajectory; it is not a new weighted-pressure theorem. The complete preceding article and historical derivations are preserved byte-for-byte in the original repository archive [3]. The physical pairing model, regular derivative estimate and spectral average estimates remain source-pinned inputs rather than consequences of finite algebraic diagnostics.

From momentum fibers to the original target roof. A geometric pressure bound on endpoint momentum fibers does not by itself control a target-roof fiber: the momentum magnitudes of its words need not agree. Theorem 9.2 removes this obstruction to summation. Convexity of the Cartesian broken-path length gives two support inequalities between any two physical trajectories of one lifted itinerary. On a common endpoint-gradient ray these inequalities order their lengths strictly by gradient magnitude. Thus a word supplies at most one point at a specified roof and Gauss angle, without requiring a connected word domain.

Theorem 10.3 combines this global fact with the retained uniform geometric pressure and optical stable-cocycle comparison. For any set S of distinct interior marks it gives

$$\left\| \frac{dF_{\#}(\prod_{i\in S} c_i^{-1}\mathbf{1}_{\Pi}\nu_R^*)}{dt} \right\|_{\infty} \le C_\star K_\star^{|S|}.$$

All original exact labels remain on the left. On the upper side, a marked full word is reconstructed from its unmarked blocks and its finitely many deleted physical step symbols. This is a weighted summation theorem; it does not insert an uncontrolled count of words or homogeneous strips. Its constants have no Gaussian m^{-2} factor, a distinction essential to its use.

Corollary 10.4 sums changing incidence witnesses by elementary symmetric polynomials. The original geometric depth thresholds have bounded sum, independently of m. Theorem 11.2 consequently pays every persistent-source trajectory with an incidence witness beyond an explicit $O(\log m)$ end layer, with normalized height $C\varepsilon(B)(m+1)^{-\eta}$. A factorial bound also pays states with $O(\log m/\log\log m)$ or more incidence witnesses. These collision-dependent classifications are substituted only into bounds valid at every finite count; all spectral windows and outer scales remain fixed in their original order.

This leaves the original pointwise target, not a new topic, at (11.12). The remaining shallow incidence and clearance source is not declared small by an unscaled pressure estimate. Sections 9–11 contain the retained roof-pressure proofs; all preceding source and capacity arguments remain in the article, with the historical packet pinned in the original repository under `archive/v111/archive/v110` [3].

Joint incidence load and the two collision-end layers. A witness at depth d_i already spends the amount αd_i of logarithmic incidence load, where $\alpha = |\log q_0|/4$. Its additional smallness spends $\log(2\varepsilon q_0^{d_i/4}/c_i)$. These contributions are nonnegative and can be summed before selecting

a witness. The resulting variable $\mathscr{L}_\varepsilon$ obeys

$$\left\| \frac{dF_{\#}(e^{\lambda \mathscr{L}_\varepsilon} \mathbf{1}_\Pi \nu_R^*)}{dt} \right\|_\infty \le C_\star \exp\left\{ \frac{4K_\star \varepsilon}{e^{\alpha(1-\lambda)} - 1} \right\}, \qquad 0 < \lambda < 1.$$

Theorem 12.1 proves this by a positive product expansion on the original physical source. Optimization gives the tail $C_\star \exp[-u + 2\sqrt{b_\varepsilon u}]$, where $b_\varepsilon = 4K_\star \varepsilon/\alpha$. At $u = (2+\eta)\log(m+1)$ this pays the normalized height, including states with only one sufficiently severe shallow witness.

The depth multiset has at most two entries at every level. Lemma 13.1 preserves this fact in the coefficient of $\prod_{d\ge 1}(1+zr^d)^2$, instead of bounding that coefficient by a factorial involving only the total threshold sum. It proves the quadratic exponent $r^{\lfloor (s+1)^2/4 \rfloor}$. The resulting Theorem 13.2 reduces the witness-count threshold to order $\sqrt{\log m}$. Sharpness is established for this depth polynomial, not claimed for an unproved physical lower bound.

Theorem 14.1 incorporates the retained incidence-load payment as a positive restriction of the original source. Its retained part has bounded total incidence depth and bounded total logarithmic severity at the displayed collision scale. Clearance witnesses, including unvisited tangency, remain distinct and may occur at every depth. The full preceding article and all of its historical material are preserved in the original repository at `archive/v111` [2]. The current article retains every preceding active proof, promotes the geometric input conventions explicitly, and supplies the unbounded-weight exhaustion and marked-step indexing used by the target-roof sum.

1. The unchanged physical problem

Put $R_- = 9/20$ and $R_+ = 47/100$, and let $R \in I = [R_-, R_+]$. Circular scatterers have centers in $\Lambda = \mathbb{Z}(1,0) + \mathbb{Z}(1/2, \sqrt{3}/2)$. Write T_R for the regular collision map and $(\alpha, p) \in (\mathbb{R}/2\pi\mathbb{Z}) \times (-1,1)$ for its coordinates; p is the outgoing tangential velocity and $\mathrm{d}\nu = \mathrm{d}\alpha\, \mathrm{d}p/(4\pi)$. Physical flights satisfy $\tau_- \le \tau_R \le \tau_+$, where $\tau_- = 3/50$ and τ_+ is the fixed finite-horizon bound.

The section is specified here to fix every event in the argument. Put $h_0 = 1/200$. Let Y_R be the union of the four open coordinate squares of half-side h_0 centered at

$$(0,0), \quad (\pi/6, -1/2), \quad (-\theta_R, \sin\theta_R), \quad (\pi/3 - \theta_R, \sin\theta_R).$$

Here $\theta_R = 2\arctan q_R$, and q_R is the unique root in $(49/100, 53/100)$ of

$$8Rq(1+q^2) - 4q(1-q^2) + \sqrt{3}(1 - 6q^2 + q^4) = 0.$$

Set

$$U = \{|\alpha - \pi/6| < 3/50,\ |p| < 3/20\}, \qquad Y_R^* = Y_R \cup U, \qquad c_* = \frac{91}{10000\pi}.$$

The rectangles are disjoint and $\nu(Y_R^*) = c_*$. We always use $\nu_R^* = c_*^{-1}\nu|_{Y_R^*}$, not a renormalization of a smaller source.

At m flights, let $L = L_{m,R}$ be their total length, $K^{\mathrm{c}}_{m,R}$ their lattice displacement, and $A_{m,R} = \sum_{j=0}^{m-1} \mathbf{1}_{Y_R^*} \circ T_R^j$. The event is

$$\Pi_{n,k,m,R} = Y_R^* \cap \{K^{\mathrm{c}}_{m,R} = k,\ A_{m,R} = n\} \cap T_R^{-m} Y_R^*. \tag{1.1}$$

The initial visit is counted; the terminal visit is tested separately. On this event the actual nth return has collision count m and roof L. Denote its density by $p_{n,R}(k,m,t)$.

Write $c_j = \sqrt{1-p_j^2}$ and let Δ_j be the least clearance of flight j from any third disk, excluding its two incident disks. Keep the original smooth function $0 \le \Theta \le 1$, zero on $(-\infty, 1]$ and one on

$[2,\infty)$. With $q_0 = 47/53$, $d_j = \min(j, m-j)$ and $d'_j = \min(j, m-1-j)$, set

$$H^\varepsilon = \prod_{j=0}^{m} \Theta\left(\frac{c_j}{\varepsilon q_0^{d_j/4}}\right) \prod_{j=0}^{m-1} \Theta\left(\frac{\Delta_j}{\varepsilon q_0^{d'_j/4}}\right), \qquad W^\varepsilon = 1 - H^\varepsilon, \tag{1.2}$$

$$b^\varepsilon(t)\,\mathrm{d}t = L_\#(W^\varepsilon \mathbf{1}_{\Pi_{n,k,m,R}} \nu_R^*). \tag{1.3}$$

In particular first incidence and first clearance outside any selected critical neighborhoods are still present. No section-distance factor has been inserted into this guard.

1.1. **The arithmetic reference and the target.** For $y = (k_1, k_2, t, n)$ put

$$\mu_R = (0, 0, \bar\tau_R, c_*), \quad Z = (y - m\mu_R)/\sqrt{m}, \quad \mathcal{C}_{m,R,n,k}(M) = \{t : |Z| \le M\}.$$

We retain the finite spectral-peak reference, including peaks close to but not on the unit circle. To specify its meaning, the collision record observable is $(\kappa_1, \kappa_2, \tau, \mathbf{1}_{Y_R^*})$. Let $\lambda_j(R,\vartheta)$ and $\mathsf{P}_j(R,\vartheta)$ be its continued simple twisted eigenvalue and projection in each of the fixed peak charts, and let χ_j be their frequency partition. At the real maximum a_j of $\log|\lambda_j|$ put

$$z_j = \lambda_j(R, a_j), \quad \mu_j = i^{-1}\nabla \log \lambda_j(R, a_j), \quad H_j = -\nabla^2 \log \lambda_j(R, a_j),$$

with $\operatorname{Re} H_j \ge aI_4$ for a common $a > 0$. If $\eta_R = \mathbf{1}_{Y_R^*}$ and ℓ denotes mass evaluation, put $B_j = \chi_j(R, a_j)\ell(M_{\eta_R}\mathsf{P}_j(R, a_j)(\eta_R \nu))$. Define

$$G_H(z) = (2\pi)^{-4} \int_{\mathbb{R}^4} e^{-iv\cdot z - v^{\mathsf{T}} H v/2}\,\mathrm{d}v,$$

$$\mathcal{L}_{m,R}(y) = \operatorname{Re}\left[c_*^{-1} \sum_j e^{-ia_j\cdot y} z_j^m B_j G_{H_j}\left((y - m\mu_j)/\sqrt{m}\right)\right]. \tag{1.4}$$

Only the charts active at R are summed. This is the same transition reference as in the retained spectral theorem. The existence and uniform estimates for these scalar data belong to that analytic input, not to the transport argument below. No frequency branch or arithmetic zero class is deleted.

The governing assertion remains

$$\sup_{R,\, 1\le n\le m,\, k\in\mathbb{Z}^2} \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} |m^2 p_{n,R}(k,m,t) - \mathcal{L}_{m,R}(k_1,k_2,t,n)| \longrightarrow 0. \tag{1.5}$$

Let $\mathcal{E}_M$ denote this error's collision limsup. An empty central target contributes zero. The order of the uniformity is $\sup_R \operatorname{ess\,sup}_t$, so exceptional density sets may depend on R.

2. The physical source and the inherited analytic inputs

We specify the inputs to the new geometric argument, separating the existence of the original physical source from the endpoint regularity that will be proved below. The complete historical proofs and their qualifications are retained in the accompanying archive.

For fixed m, R, n, k, let Θ_{phys} be the original disjoint incident-component pairing-sheet space. It is the countable Borel sum of the maximal regular components of the original exact event, parametrized by

$$(u,\theta) = (F, \arg\nabla_q F), \qquad q = (\alpha_0, \alpha_m), \qquad a(\theta) < u < b(\theta), \quad \ell = b - a > 0.$$

The sheet identity, and not just its angular coordinate, is included in θ. Integration $d\theta$ means the original one-copy physical sum with angular Lebesgue measure. The regular components, their Borel endpoint maps, their physical multiplicity convention, and their common coarea versions

are the retained pairing-space input. This article does not replace them by symbolic collision words. Every component carries the same exact event (1.1). Put

$$\begin{aligned} g &= \nabla_q F = R(-p_0, p_m), \quad z = |g|, \quad I_m = 1 + \sum_{i=1}^{m-1} q_0^{d_i}/c_i, \\ \omega &= |\partial_{p_0}\alpha_m|^{-1}, \quad \chi = |\det D_q(F, \arg g)|, \quad A = \frac{\omega}{4\pi c_* \chi}, \quad f^\varepsilon = AW^\varepsilon. \end{aligned} \tag{2.1}$$

The density formula is used initially on the regular interior. Once its endpoint traces have been established, the original pairing is

$$f_-^\varepsilon = f^\varepsilon(a+,\theta), \quad f_+^\varepsilon = f^\varepsilon(b-,\theta), \quad c^\varepsilon = \min(f_-^\varepsilon, f_+^\varepsilon), \quad d\pi^{\varepsilon,+} = c^\varepsilon \mathbf{1}_{\{a<u<b\}} d\theta\, du. \tag{2.2}$$

Write $p_\vee = \max(f_-^\varepsilon, f_+^\varepsilon)$ and $e_\partial = |f_-^\varepsilon - f_+^\varepsilon|$. Thus $p_\vee = c^\varepsilon + e_\partial$.

Input statement 2.1 (Regular variation and completed physical averages)**.** The inherited regular-point optical calculation gives, almost everywhere on each regular segment,

$$|\partial_u f^\varepsilon| \le C(1+\varepsilon^{-1}) A I_m / z. \tag{2.3}$$

Let $\mathsf{T}_\Delta^\varepsilon$ be the complete paired roof density restricted to $0 < \ell < \Delta$, with no additional selector, and let

$$v^0(t)\, dt = F_\#(I_m z^{-1} \mathbf{1}_\Pi \nu_R^*), \qquad A_r f(t) = (2r)^{-1} \int_{t-r}^{t+r} f(u)\, du.$$

For every previously fixed $\varepsilon, r, \Delta, \rho > 0$,

$$\limsup_m \sup_{R,n,k} m^2 \|A_r \mathsf{T}_\Delta^\varepsilon\|_\infty \le C\{\varepsilon^{1/16} + (1+\varepsilon^{-1})\Delta\}, \tag{2.4}$$

$$\limsup_m \sup_{R,n,k} m^2 \|A_\rho v^0\|_\infty \le C. \tag{2.5}$$

The constants are independent of the previously fixed widths. The coarea identity identifies the full physical sum of AI_m/z with v^0. Fixed-count local finiteness and common roof versions are retained. These are averaged, not pointwise, bounds.

These are the positive pairing, uncut volume and weighted optical variation statements in Section 2 of the frozen article [5]. Their exact historical source labels are recorded in the proof ledger. The new proof uses (2.3) only on regular interiors and supplies its integrable passage to the physical endpoints.

2.1. **The original microscopic source.** The outer schedule is

$$\varepsilon(B) = A_0 B^{-1/12}, \qquad r_0 = c_0^{\mathrm{out}} \varepsilon(B)^4, \qquad \delta(B) = \tfrac{3}{4} r_0, \qquad \Delta_B = 8\delta(B). \tag{2.6}$$

The fixed constants A_0, c_0^{out} are the inherited constants; c_0^{out} is written this way to avoid confusion with the initial incidence c_0. Keep the original remainder selector H_B, the paid critical-arm family $\mathscr{C}_B$, and the adaptive paid family $\mathscr{D}_B$ of the historical microscopic reduction. Their memberships are Borel, disjointized by first physical membership, and their total outer costs are at most $C\varepsilon(B)^{-1/32}$. For a previously fixed $0 < h_B < \delta(B)$ put

$$\begin{aligned} \widetilde{H}_B &= H_B \mathbf{1}_{(\mathscr{C}_B \cup \mathscr{D}_B)^c} \mathbf{1}_{\{0<\ell<h_B\}}, \\ \mathsf{U}_B^{\mathrm{micro}}(t) &= \int c^{\varepsilon(B)} \mathbf{1}_{\{a<t<b\}} \widetilde{H}_B(\Psi(\theta,t))\, d\theta. \end{aligned} \tag{2.7}$$

No sign cancellation, deletion of an endpoint class, or change of section occurs in this definition. Here Ψ denotes the original physical parametrization.

Input statement 2.2 (The scalar microscopic reduction)**.** For every fixed M, define

$$\mathcal{H}_M(T) = \limsup_m \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} T(t).$$

The historical exact positive reduction gives

$$\mathcal{E}_M \le \mathcal{H}_M(\mathsf{U}_B^{\mathrm{micro}}) + C_M B^{-1/384}. \tag{2.8}$$

The assertion (1.5) is equivalent to $\mathcal{H}_M(\mathsf{U}_B^{\mathrm{micro}}) \to 0$ for every fixed M. All widths and costs are fixed at the outer stage before the complete physical sum, the essential suprema, and the collision limsup; B is outermost.

This is the unchanged theorem in historical module 255. The fixed-positive-residence estimate in module 253 and the earlier paid-arm estimates are retained as its inputs. A bound uniform in a previously fixed positive residence is not used at a collision-dependent residence.

3. The reduced optical action

Fix an actual collision itinerary with $m \ge 1$ flights. Its visited centres are $O_0, \dots, O_m$, its collision points are $x_i = O_i + Rn(\alpha_i)$, and $n(\alpha) = (\cos\alpha, \sin\alpha)$. Write $\tau_i = |x_{i+1} - x_i|$. All variations in this section are made on this actual itinerary before a physical boundary is reached. They neither continue an event through a forbidden collision nor change an occupation label.

Lemma 3.1 (The original section stays away from endpoint grazing)**.** *On the closure of the original initial and terminal section tests,*

$$|p_0|, |p_m| \le \frac{9}{10}, \qquad c_0, c_m \ge c_\circ := \frac{\sqrt{19}}{10}. \tag{3.1}$$

Proof. The only section centres with tangential coordinate exceeding $1/2$ in absolute value have coordinate $2q_R/(1+q_R^2)$. This function is increasing on $(0,1)$, and

$$\frac{2(53/100)}{1+(53/100)^2} + \frac{1}{200} < \frac{9}{10}.$$

The other squares and the rectangle U satisfy the same strict bound. Passing to their closures proves the assertion. □

Let $S(\alpha_0, \dots, \alpha_m) = \sum_{i=0}^{m-1} \tau_i$ be the broken-path action. At a regular specular trajectory its interior variables are stationary. Eliminating them locally gives the actual endpoint action $F(q)$, where $q = (\alpha_0, \alpha_m)$.

Lemma 3.2 (Second variation and its Schur complement)**.** *For a real variation $\xi = (\xi_0, \dots, \xi_m)$ at an actual regular trajectory,*

$$\begin{aligned} D^2S[\xi,\xi] = R\left(c_0\xi_0^2 + c_m\xi_m^2 + 2\sum_{i=1}^{m-1} c_i\xi_i^2\right) \\ + R^2 \sum_{i=0}^{m-1} \frac{(c_i\xi_i + c_{i+1}\xi_{i+1})^2}{\tau_i}. \end{aligned} \tag{3.2}$$

The reduced Hessian $G = D_q^2 F$ satisfies

$$R\,\mathrm{diag}(c_0, c_m) \le G, \qquad \lambda_\circ I_2 \le G \le \Lambda_\circ I_2, \quad \lambda_\circ = \frac{9\sqrt{19}}{200}, \quad \Lambda_\circ = \frac{47}{6}. \tag{3.3}$$

These inequalities are uniform in the radius, the collision count, and the original exact labels.

Proof. For a flight with unit direction v, the Hessian of its length in its endpoints is the squared perpendicular displacement divided by the flight length. The acceleration of $Rn(\alpha)$ is $-Rn(\alpha)$. At the departure and arrival points, respectively, $v\cdot n_i = c_i$ and $v\cdot n_{i+1} = -c_{i+1}$. Also $(Rn_i')\cdot v^{\perp} = Rc_i$ and $(Rn_{i+1}')\cdot v^{\perp} = -Rc_{i+1}$. Thus one flight contributes $R(c_i\xi_i^2 + c_{i+1}\xi_{i+1}^2)$ and the corresponding square in (3.2). Summing proves that formula.

Let $H = D^2S$, $E = \{0, m\}$ and $I = \{1, \dots, m-1\}$. For $m > 1$, the interior block is positive definite. The implicit function theorem and differentiation of the stationary action give

$$G = H_{EE} - H_{EI}H_{II}^{-1}H_{IE}, \qquad \eta^{\mathsf{T}}G\eta = \min_{\zeta}(\eta,\zeta)^{\mathsf{T}}H(\eta,\zeta). \tag{3.4}$$

The order in the last expression places endpoint coordinates first. The diagonal endpoint terms in (3.2) prove the first lower bound. Choosing the interior variation to be zero gives $G \le H_{EE}$. The latter block is bounded above by $\{R_+ + 2R_+^2/\tau_-\}I_2 = (47/6)I_2$; the same bound covers the single-flight case, where there is no Schur complement. Lemma 3.1 proves the remaining lower bound. □

3.1. **A positive optical denominator.** Put

$$\mathsf{S}(s) = \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix}, \qquad \mathsf{C}(\kappa) = \begin{pmatrix} 1 & 0 \\ \kappa & 1 \end{pmatrix}, \qquad \kappa_i = \frac{2}{Rc_i}. \tag{3.5}$$

Let M_m be the product from the initial collision to the terminal collision, with the rightmost factor acting first:

$$\mathsf{M}_m = \mathsf{S}(\tau_{m-1})\mathsf{C}(\kappa_{m-1})\cdots\mathsf{S}(\tau_1)\mathsf{C}(\kappa_1)\mathsf{S}(\tau_0), \qquad \mathfrak{b}_m = (\mathsf{M}_m)_{12}. \tag{3.6}$$

For $m = 1$ the product is just $\mathsf{S}(\tau_0)$.

Lemma 3.3 (Exact denominator and a marked incidence bound)**.** *The actual endpoint-coordinate density is*

$$\omega = |\partial_{p_0}\alpha_m|^{-1} = \frac{|G_{0m}|}{R} = \frac{Rc_0c_m}{\mathfrak{b}_m}. \tag{3.7}$$

With $\varrho = 47/59$, one has

$$\omega \le \frac{47}{6}\varrho^{m-1}, \qquad \frac{\omega}{c_j} \le \frac{2209}{72}\varrho^{m-2} \quad (1 \le j \le m-1). \tag{3.8}$$

The second inequality is asserted when $m \ge 2$.

Proof. The first equality with G follows by differentiating $p_0 = -\partial_{\alpha_0}F/R$ with α_0 fixed. For the denominator, the interior block of H is $R^2C_IJC_I$, where $C_I = \operatorname{diag}(c_1, \dots, c_{m-1})$ and J is tridiagonal with

$$J_{ii} = \kappa_i + \tau_{i-1}^{-1} + \tau_i^{-1}, \qquad J_{i,i+1} = \tau_i^{-1}.$$

The corner cofactor in (3.4) gives $|G_{0m}| = R^2c_0c_m/(\prod_i \tau_i \det J)$. Expansion of successive tridiagonal determinants gives $\prod_i \tau_i \det J = \mathfrak{b}_m$. This identity also follows by multiplying the second column in (3.6); both recursions start with $\mathfrak{b}_1 = \tau_0$. The empty determinant convention handles $m = 1$.

All entries in the optical factors are nonnegative. Multiplication of a positive second column $(b, d)^{\mathsf{T}}$ by $\mathsf{S}(\tau_i)\mathsf{C}(\kappa_i)$ changes its top entry to $(1+\tau_i\kappa_i)b+\tau_i d$. Since $1+\tau_i\kappa_i \ge 1+2\tau_-/R_+ = 59/47$,

$$\mathfrak{b}_m \ge \tau_-\varrho^{-(m-1)}. \tag{3.9}$$

For a fixed interior collision j, split the product immediately before and after $\mathsf{C}(\kappa_j)$. Its $(1,2)$ entry is an expression with nonnegative coefficients in κ_j, whose linear coefficient is $\mathfrak{b}_{\text{right}}\mathfrak{b}_{\text{left}}$. The two subproducts have respectively $m-j$ and j flights. Hence

$$\mathfrak{b}_m \ge \kappa_j\tau_-^2\varrho^{-(m-2)}.$$

Substitution into (3.7), and $c_0, c_m \le 1$, proves (3.8). □

Corollary 3.4 (The full inherited incidence insertion)**.** *For the unchanged insertion* $I_m = 1 + \sum_{i=1}^{m-1} q_0^{d_i}/c_i$, *with* $q_0 = 47/53$,

$$\omega I_m \le 500\varrho^{(m-2)_+}. \tag{3.10}$$

This is an estimate on each actual physical trajectory; it involves no lower cutoff on any interior incidence.

Proof. For $m \ge 2$, $\sum_{i=1}^{m-1} q_0^{d_i} \le 2q_0/(1-q_0) = 47/3$. Use (3.8) and $(47/6)\varrho + (2209/72)(47/3) < 500$. For $m = 1$ the sum is empty and the first optical bound suffices. □

No estimate for a sum over collision words is obtained by simply summing the right side of (3.10). In particular, this per-trajectory exponential estimate is not being substituted for a collision-uniform local limit theorem.

4. Completion of the original roofwise endpoint traces

We use the original incident-component pairing sheets and their constant-angle coordinates $(u, \theta) = (F, \arg \nabla F)$. On one of their positive-length maximal regular segments, write $a < u < b$, $g = \nabla F = ze(\theta)$ and $e^{\perp}(\theta) = (-\sin\theta, \cos\theta)$. The physical section and all original exact-event tests are retained.

Lemma 4.1 (Finite Gauss-coordinate length)**.** *Along a fixed-*θ *segment,*

$$\frac{dq}{dz} = G^{-1}e, \qquad \frac{du}{dz} = ze^{\mathsf{T}}G^{-1}e, \qquad 2\lambda_\circ \le \frac{d(z^2)}{du} \le 2\Lambda_\circ. \tag{4.1}$$

Consequently z *is strictly increasing, both endpoint limits of* q *exist, and*

$$\int_a^b \frac{du}{z(u,\theta)} \le \lambda_\circ^{-1}(z_b - z_a) \le \lambda_\circ^{-1} \min\{\sqrt{2}R_+, \sqrt{2\Lambda_\circ(b-a)}\}. \tag{4.2}$$

A critical endpoint of a positive-length segment can only be its incoming endpoint.

Proof. Differentiate $\nabla F(q) = ze$ and then $F(q) = u$. The bounds in (3.3) give (4.1). They also give $|dq/dz| \le \lambda_\circ^{-1}$. Since $0 < z \le \sqrt{2}R_+$, the two endpoint limits of q exist in the chosen angular lifts. Integrating $du/z = e^{\mathsf{T}}G^{-1}e\,dz$ proves the first inequality in (4.2). The second uses $z_b^2 - z_a^2 \le 2\Lambda_\circ(b-a)$ and $(z_b - z_a)^2 \le z_b^2 - z_a^2$. If $z_b = 0$, strict increase would contradict $z > 0$ in the interior. □

Lemma 4.2 (A continuous reduced Hessian at visited grazing)**.** *The reduced Hessian has a continuous extension along a fixed itinerary when any collection of its interior* c_i *tends to zero, provided* $\tau_i \ge \tau_-$ *and* (3.1) *hold. Its extension still satisfies* (3.3)*. If at least one interior* c_j *vanishes, the extended mixed entry* G_{0m} *vanishes.*

Proof. Let $L = \sum_{i=0}^{m-1} \tau_i^{-1}(e_i + e_{i+1})(e_i + e_{i+1})^{\mathsf{T}}$, and set $C_E = \mathrm{diag}(c_0, c_m)$. For $m > 1$ put

$$Z = 2I + RC_I^{1/2}L_{II}C_I^{1/2} \ge 2I.$$

Formula (3.4) can be written without inverse incidences:

$$\begin{aligned} G = {} & RC_E + R^2C_EL_{EE}C_E \\ & - R^3C_EL_{EI}C_I^{1/2}Z^{-1}C_I^{1/2}L_{IE}C_E. \end{aligned} \tag{4.3}$$

Every term is continuous for $c_i \ge 0$, since Z^{-1} remains bounded. The order inequalities pass to this limit. The marked bound in (3.8), together with $|G_{0m}| = R\omega$, proves the last assertion. There is no interior extension to make when $m = 1$. □

At an endpoint of a fixed-θ segment the complete finite trajectory also has a limit. Indeed Lemma 4.1 gives the initial angle and $p_0 = -z\cos\theta/R$. For each designated next disk, the entry root of the flight quadratic is continuous up to a tangency. The flight length cannot collapse to zero, and the reflection $v \mapsto v - 2(v\cdot n)n$ is continuous at tangency. Induction over the fixed finite itinerary gives the limits of all visited points, velocities, τ_i and c_i. The third-disk clearances have limits as well: the finite horizon restricts the possible nearby disk centres to a finite set, and the minimum distance to each of them is continuous. This argument takes only one-sided limits from the actual event; it asserts no continuation through its boundary.

Theorem 4.3 (Trace completion on the original physical sheets)**.** *On every positive-length original regular pairing segment the coarea factor and weighted density are*

$$\chi = \det D_q(F, \arg g) = e^{\perp\mathsf{T}} G e^{\perp} \geq \lambda_\circ, \qquad f^\varepsilon = \frac{\omega W^\varepsilon}{4\pi c_* \chi}. \tag{4.4}$$

They have finite one-sided endpoint limits. The regular-point variation input (2.3) *implies that f^ε extends absolutely continuously to the closed segment and satisfies*

$$|f^\varepsilon(b-) - f^\varepsilon(a+)| \leq C(1+\varepsilon^{-1}) \int_a^b AI_m/z \, du. \tag{4.5}$$

In fact, on each physical sheet,

$$|f^\varepsilon(v) - f^\varepsilon(u)| \leq C(1+\varepsilon^{-1})\varrho^{(m-2)_+}|v-u|^{1/2} \quad (a \leq u < v \leq b). \tag{4.6}$$

If a limiting trajectory at either endpoint contains an actually visited interior grazing collision, its trace at that endpoint is zero. The original paired minimum on that entire segment is then zero, for every original selector.

Proof. The determinant identity in (4.4) follows by expressing $D\arg g$ as $e^{\perp\mathsf{T}} G/z$. The preceding endpoint-limit argument and Lemma 4.2 give limits for G, ω and χ, the last with a positive lower bound. Every factor in the finite product defining W^ε is continuous in its incidence or clearance argument. Thus f^ε has the asserted finite limits, including at a critical incoming endpoint, where the direction θ is fixed.

Corollary 3.4 and (4.4) bound AI_m by $C\varrho^{(m-2)_+}$. The integral in (4.5) is finite by Lemma 4.1. On compact regular subsegments the flight charts are smooth and the finite clearance minima are locally Lipschitz, so f^ε is locally absolutely continuous. Integrating the inherited regular-point derivative inequality on these subsegments and then passing to the proved endpoint limits establishes absolute continuity and (4.5). The same argument between u and v, using (4.2), gives (4.6).

Finally, if $c_j \to 0$ at a visited interior collision, the marked optical bound gives $\omega \to 0$, while $0 \leq W^\varepsilon \leq 1$ and $\chi \geq \lambda_\circ$. Hence that endpoint trace is zero. The paired density is its nonnegative minimum with the other trace. □

Remark 4.4 (Two different tangency mechanisms)*.* The last assertion concerns a collision in the visited itinerary. A flight which merely becomes tangent to an *unvisited* third disk has zero clearance but need not have any $c_j = 0$. If all visited incidences remain positive, then (3.7) has a positive limit. A zero third-disk clearance makes $W^\varepsilon = 1$ at the limit; consequently the trace can be positive. Such sheets are not removed. Neither are nongrazing critical entrances or positive section-boundary traces. For a nongrazing critical entrance the exact directional limit is $\omega_* W_*^\varepsilon/(4\pi c_* e^{\perp\mathsf{T}} G_* e^{\perp})$. No roof-independent value at the critical point is asserted.

5. The complete physical comparison space

The comparison space is now specified without an unverified endpoint-trace exclusion:

$$\Theta_{\mathrm{ad}} = \Theta_{\mathrm{phys}}. \tag{5.1}$$

More explicitly, it consists of all positive-length maximal regular pairing segments of the original exact event, with their original incident-component identities. Their endpoints may be critical, visited-grazing, third-disk-tangent, or section-boundary limits. Theorem 4.3 supplies finite traces and the endpoint variation inequality on this space. The underlying Borel pairing construction and the averaged analytic estimates remain the inputs in Section 2. No singular trajectory is continued past its physical boundary, and no new source supported at a zero-length segment is introduced.

For fixed $0 < r < \Delta$, define

$$\begin{aligned} q_+(t,\theta) &= \mathbf{1}_{\{0<\ell<\Delta,\ a<t,\ t+r<b\}},\\ q_-(t,\theta) &= (1-q_+)\mathbf{1}_{\{0<\ell<\Delta,\ a<t-r,\ t<b\}}, \qquad Q_r = q_+ + q_-. \end{aligned} \tag{5.2}$$

Thus Q_r is Borel and at most one. Write

$$P_r(t) = \int Q_r p_\vee \, d\theta, \qquad D_r(t) = \int Q_r c^{\varepsilon} \, d\theta, \qquad E_r(t) = \int Q_r e_\partial \, d\theta.$$

The full pool is retained even on a sheet whose paired source is zero.

Theorem 5.1 (Capacity on the full original comparison space)**.** *At finite collision count, on the original common roof versions,*

$$P_r(t) \le \frac{1}{r}\int_{t-r}^{t+r} \mathsf{T}^{\varepsilon}_{\Delta}(u)\, du + C(1+\varepsilon^{-1})\int_{t-\Delta}^{t+\Delta} v^0(u)\, du. \tag{5.3}$$

Consequently, for every previously fixed $K > 0$,

$$\mathcal{H}_M(KP_r) \le CK\{\varepsilon^{1/16} + (1+\varepsilon^{-1})\Delta\}. \tag{5.4}$$

Proof. Every positively oriented selected segment contains $[t, t+r]$; every negatively oriented one contains $[t-r, t]$. Integrating its constant paired minimum over the appropriate window gives r times that minimum. Tonelli and $Q_r \le 1$ therefore give $D_r(t) \le r^{-1}\int_{t-r}^{t+r} \mathsf{T}^{\varepsilon}_{\Delta}$. Each selected segment lies inside $(t-\Delta, t+\Delta)$. The completed variation inequality (4.5) and the original one-copy physical coarea sum give $E_r(t) \le C(1+\varepsilon^{-1})\int_{t-\Delta}^{t+\Delta} v^0$. Since $P_r = D_r + E_r$, this proves (5.3). Use (2.4) and (2.5), only on these previously fixed windows, to obtain (5.4). The weights are held at the original roof t throughout; none is differentiated or evaluated at the integration roof u. □

5.1. **The physical grazing partition.** For each interior collision let c_i^- and c_i^+ be its incidence limits at the two ends of a pairing segment. Define the Borel sets

$$\Gamma = \bigcup_{i=1}^{m-1} \{c_i^- = 0 \text{ or } c_i^+ = 0\}, \qquad \mathcal{N} = \Theta_{\mathrm{phys}} \setminus \Gamma. \tag{5.5}$$

For $m = 1$ the union is empty. The limits are Borel, for example by evaluating the interior functions at $a + \ell/j$ and $b - \ell/j$ and letting the integer j tend to infinity. Set

$$P_r^{\mathrm{gr}} = \int_{\Gamma} Q_r p_\vee \, d\theta, \qquad P_r^{\mathrm{ng}} = \int_{\mathcal{N}} Q_r p_\vee \, d\theta, \qquad P_r = P_r^{\mathrm{ng}} + P_r^{\mathrm{gr}}.$$

The superscripts refer to visited collision incidences in the limiting finite itinerary, not to a lower incidence cutoff.

Theorem 5.2 (Zero original grazing source and paid grazing capacity)**.** *For every original Borel selector $0 \le H \le 1$, at finite count,*

$$\int_\Gamma c^\varepsilon \mathbf{1}_{\{a<t<b\}} H(\Psi(\theta,t))\, d\theta = 0. \tag{5.6}$$

Nevertheless P_r^{gr} need not vanish. It satisfies

$$P_r^{\mathrm{gr}}(t) \le C(1+\varepsilon^{-1}) \int_{t-\Delta}^{t+\Delta} v^0(u)\, du, \tag{5.7}$$

$$\mathcal{H}_M(KP_r^{\mathrm{gr}}) \le CK(1+\varepsilon^{-1})\Delta. \tag{5.8}$$

These estimates include the complete positive capacity on this physical class, with no count of angular blocks or collision words.

Proof. Theorem 4.3 gives $c^\varepsilon = 0$ on Γ, proving (5.6) before the roof pushforward. On that same class $p_\vee = e_\partial$, not necessarily zero. Apply (4.5), use the localization $(a,b) \subset (t-\Delta, t+\Delta)$ on $\{Q_r = 1\}$, and complete the one-copy positive physical sum. This proves (5.7). Equation (2.5) at the fixed width Δ proves (5.8). □

6. The original microscopic shortage after grazing payment

All allocations below are scalar allocations at the original roof; they are not couplings of individual source and comparison sheets. For any original source $U(t)$ define

$$V_{r,K}(t) = \min\{U(t), KP_r(t)\}, \qquad R_{r,K}(t) = (U(t) - KP_r(t))_+.$$

The multiplier is $\beta_{r,K} = \min\{1, KP_r/U\}$ on $U > 0$ and zero on $U = 0$. Then $U = V_{r,K} + R_{r,K}$ at finite count and $V_{r,K}$ is the original source multiplied by $\beta_{r,K}(t)$. When a restricted comparison space is used, source outside it is still in U and may receive this same global fraction. It is not therefore retained sheetwise with coefficient one.

Use the exact source (2.7) and, for a concrete outer choice, put

$$h_B = \delta(B)/2, \qquad r_B = \delta(B)/8, \qquad K_B = \max\{1, \varepsilon(B)^{-1/32}\}. \tag{6.1}$$

For all sufficiently large B these satisfy the inherited cost budget. Define

$$\begin{aligned} \mathsf{R}_B^{\mathrm{full}} &= (\mathsf{U}_B^{\mathrm{micro}} - K_B P_{r_B})_+, \\ \mathsf{R}_B^{\mathrm{ng}} &= (\mathsf{U}_B^{\mathrm{micro}} - K_B P_{r_B}^{\mathrm{ng}})_+. \end{aligned} \tag{6.2}$$

The full capacity pool includes any sheets previously excluded solely for lack of a trace verification. On the complete comparison space it is exactly the canonical pool; on a previously restricted space it is an enlargement, not an identification of different capacities.

Theorem 6.1 (Quantitative payment of the visited-grazing class)**.** *Under the retained physical-source and analytic inputs, for the original event and every fixed M,*

$$0 \le \mathsf{R}_B^{\mathrm{ng}} - \mathsf{R}_B^{\mathrm{full}} \le K_B P_{r_B}^{\mathrm{gr}}, \tag{6.3}$$

$$\mathcal{H}_M(\mathsf{R}_B^{\mathrm{ng}} - \mathsf{R}_B^{\mathrm{full}}) \le C_M B^{-95/384}, \tag{6.4}$$

$$\mathcal{E}_M \le \mathcal{H}_M(\mathsf{R}_B^{\mathrm{ng}}) + C_M B^{-1/384}. \tag{6.5}$$

The unchanged pointwise assertion (1.5) *is equivalent to $\mathcal{H}_M(\mathsf{R}_B^{\mathrm{ng}}) \to 0$ for every fixed M. The original microscopic source in these formulas is supported on $\mathcal{N}$ without a change of its positive weight.*

Proof. For $u, a, b \geq 0$, the positive-part function gives $0 \leq (u-a)_+ - (u-a-b)_+ \leq b$. Apply this with $a = K_B P^{\mathrm{ng}}_{r_B}$ and $b = K_B P^{\mathrm{gr}}_{r_B}$ to prove (6.3) at finite count. Theorem 5.2, $\Delta_B = O(\varepsilon^4)$ and $K_B = O(\varepsilon^{-1/32})$ give

$$\mathcal{H}_M(K_B P^{\mathrm{gr}}_{r_B}) \leq C\varepsilon^{-1/32}(1+\varepsilon^{-1})\varepsilon^4 = O(\varepsilon^{95/32}) = O(B^{-95/384}).$$

This proves (6.4) in the prescribed order. By Theorem 5.1, the accepted full-pool source has height at most

$$C\varepsilon^{-1/32}\{\varepsilon^{1/16} + O(\varepsilon^3)\} = O(B^{-1/384}).$$

Insert its exact positive decomposition into (2.8). Since $\mathsf{R}^{\mathrm{full}}_B \leq \mathsf{R}^{\mathrm{ng}}_B$, this proves (6.5) and sufficiency of the displayed residual limit. Necessity follows from Input statement 2.2 and $0 \leq \mathsf{R}^{\mathrm{ng}}_B \leq \mathsf{U}^{\mathrm{micro}}_B$. Equation (5.6) proves the final assertion. □

This theorem supplies a physical component of the required scarcity estimate, rather than a new certificate class. It removes an entire visited-grazing source class exactly and pays the loss of all its comparison capacity by a faster-decaying original optical volume. It does not yet bound $\mathcal{H}_M(\mathsf{R}^{\mathrm{ng}}_B)$ by a quantity tending to zero. In $\mathcal{N}$ every visited endpoint incidence is positive at each fixed count, but its infimum may tend to zero with the count. Third-disk tangencies, nongrazing critical entrances, section-boundary cuts, flat and short-reach segments, and the original comparison shortages all remain with their full weight. A collision-uniform estimate on that remaining source is still required for (1.5). Neither finite-count exhaustion nor the per-itinerary exponential bound is used in its place.

7. A positive payment from the variation on a physical segment

The comparison pool of Section 5 is left unchanged. We first make a different use of the completed endpoint estimate: it pays part of the *original source*, whether or not that source has a comparison window. All statements in this section are made on the inherited one-copy physical pairing space.

Write C_{reg} for a fixed constant in (2.3). For fixed $\varepsilon > 0$ define

$$\begin{aligned} \mathcal{V}^{\varepsilon}(\theta) &= \mathrm{Var}_{[a(\theta),b(\theta)]}\, f^{\varepsilon}(\,\cdot\,,\theta), \\ \mathcal{J}(\theta) &= \int_{a(\theta)}^{b(\theta)} \frac{A(u,\theta)I_m(u,\theta)}{z(u,\theta)}\, du. \end{aligned} \tag{7.1}$$

Endpoint values in the variation are the one-sided physical traces, not values assigned on a continuation beyond the original event.

Lemma 7.1 (Measurability and the full variation bound)**.** *Both functions in* (7.1) *are Borel and finite on every positive-length physical segment. Moreover,*

$$\mathcal{V}^{\varepsilon} = \int_a^b |\partial_u f^{\varepsilon}|\, du \leq C_{\mathrm{reg}}(1+\varepsilon^{-1})\mathcal{J}. \tag{7.2}$$

This is a bound for total variation, not merely for the difference of the two endpoint values.

Proof. Pull the closed segment back to $[0,1]$ by $u = a + s(b-a)$, using the endpoint traces at $s = 0, 1$. The interior evaluations are jointly Borel. At $s = 0$ the completed value is the limit of evaluations at $a + \ell/j$, $j \geq 2$; at $s = 1$ it is the limit of evaluations at $b - \ell/j$. The completed endpoint values, and hence the function on the closed product, are jointly Borel. The resulting function is continuous in s. Its variation is the supremum of the sums of absolute increments over finite rational partitions containing 0 and 1. Continuity permits this countable family of partitions, so its supremum is Borel. The integral defining $\mathcal{J}$ is Borel by the same change of variables and nonnegative integration. Theorem 4.3 gives absolute continuity on the closed

segment. The variation of an absolutely continuous real function is the integral of the absolute value of its derivative. Integrate (2.3); its right side is integrable by Corollary 3.4 and Lemma 4.1. This proves the equality, inequality and finiteness asserted above. □

For $\gamma \geq 1$, split the original paired coefficient into

$$c^{\varepsilon,\gamma}_{\mathrm{var}} = \min\{c^{\varepsilon}, \gamma \mathcal{V}^{\varepsilon}\}, \qquad c^{\varepsilon,\gamma}_{\mathrm{per}} = (c^{\varepsilon} - \gamma \mathcal{V}^{\varepsilon})_{+}. \tag{7.3}$$

Thus $c^{\varepsilon} = c^{\varepsilon,\gamma}_{\mathrm{var}} + c^{\varepsilon,\gamma}_{\mathrm{per}}$, with both terms nonnegative. The subscript per will be justified by a physical persistence statement in the next section. It denotes a positive *submeasure* of the original source, not a replacement of its coefficient by a different normalization.

For a jointly Borel selector $0 \leq H \leq 1$ supported on $0 < \ell < h$, set

$$T^{\varepsilon,\gamma}_{H,j}(t) = \int c^{\varepsilon,\gamma}_{j}(\theta) \mathbf{1}_{\{a<t<b\}} H(\Psi(\theta,t))\, d\theta, \qquad j \in \{\mathrm{var}, \mathrm{per}\}. \tag{7.4}$$

The original selected paired source is their exact sum. The selector is evaluated at t and is not moved to another roof in the proof.

Theorem 7.2 (A summed payment of the positive short source)**.** *At every finite collision count, on the original common roof versions,*

$$0 \leq T^{\varepsilon,\gamma}_{H,\mathrm{var}}(t) \leq C_{\mathrm{reg}} \gamma (1 + \varepsilon^{-1}) \int_{t-h}^{t+h} v^{0}(u)\, du. \tag{7.5}$$

Consequently, for every $\varepsilon, h > 0$ and $\gamma \geq 1$ fixed before the collision limit,

$$\mathcal{H}_{M}(T^{\varepsilon,\gamma}_{H,\mathrm{var}}) \leq C \gamma (1 + \varepsilon^{-1}) h. \tag{7.6}$$

The constants are uniform in the collision count, radius, exact labels, segment length and selector. No positive lower bound for ℓ or for an interior incidence is required.

Proof. For a segment meeting t and satisfying $\ell < h$, one has $(a, b) \subset (t - h, t + h)$. By (7.3) and (7.2), its selected contribution is at most

$$C_{\mathrm{reg}} \gamma (1 + \varepsilon^{-1}) \mathbf{1}_{\{a<t<b,\, \ell<h\}} H(\Psi(\theta,t)) \int_{a}^{b} A I_{m}/z\, du.$$

First sum these nonnegative quantities over the physical sheets. Tonelli's theorem applies, without an estimate for the number of sheets. At an integration roof u, each retained pair (θ, u) is still a point of the same original exact event. Dropping its selector, which is between zero and one, and using physical multiplicity one therefore bounds the sum by

$$C_{\mathrm{reg}} \gamma (1 + \varepsilon^{-1}) \int_{t-h}^{t+h} \int_{\{a(\theta)<u<b(\theta)\}} A(u,\theta) I_{m}(u,\theta)/z(u,\theta)\, d\theta\, du.$$

The coarea identity in Input statement 2.1 identifies the inner integral with $v^{0}(u)$. This proves the finite-count inequality. Apply (2.5) at the previously fixed width h, using $\int_{t-h}^{t+h} v^{0} = 2h A_{h} v^{0}$. Only after the complete sum and this inequality are established are the uniform norms and collision limsup taken. This proves (7.6). □

Corollary 7.3 (The original microscopic schedule)**.** *Keep $\widetilde{H}_{B}, h_{B}, r_{B}, K_{B}$ exactly as in* (2.7) *and* (6.1)*, and choose*

$$\gamma_{B} = \max\{2, \varepsilon(B)^{-1/32}\}. \tag{7.7}$$

Let $\mathsf{U}^{\mathrm{var}}_{B}$ and $\mathsf{U}^{\mathrm{per}}_{B}$ denote the two sources in (7.4) *with $H = \widetilde{H}_{B}$, $\varepsilon = \varepsilon(B)$ and $\gamma = \gamma_{B}$. Then*

$$\mathsf{U}^{\mathrm{micro}}_{B} = \mathsf{U}^{\mathrm{var}}_{B} + \mathsf{U}^{\mathrm{per}}_{B}, \qquad \mathcal{H}_{M}(\mathsf{U}^{\mathrm{var}}_{B}) \leq C_{M} B^{-95/384}. \tag{7.8}$$

Proof. The identity holds before pushforward, by (7.3). For sufficiently large B, $h_B = O(\varepsilon^4)$ and $\gamma_B = O(\varepsilon^{-1/32})$. Theorem 7.2 gives the bound

$$C\gamma_B(1+\varepsilon^{-1})h_B = O(\varepsilon^{95/32}) = O(B^{-95/384}).$$

All three quantities $h_B, \varepsilon(B), \gamma_B$ are fixed before m tends to infinity. No collision-dependent residence is inserted into an estimate asserted only for fixed residence. □

Unlike (5.6), this payment does not rely on a zero paired coefficient. If $c^\varepsilon > 0$ and $\mathcal{V}^\varepsilon > 0$, its coefficient in (7.3) is positive. If $c^\varepsilon \le \gamma\mathcal{V}^\varepsilon$, the complete original coefficient is paid. These statements do not assert a positive uniform proportion of paid mass in every exact event.

8. Persistent guarded segments and the same pointwise target

The positive decomposition of the preceding section has a geometric consequence on every segment which retains source. It uses the original incidence and clearance guard, without adding a distance to the section boundary.

Theorem 8.1 (Persistence of the guard on the remaining source)**.** *Let $\gamma > 1$. If $c^{\varepsilon,\gamma}_{\mathrm{per}}(\theta) > 0$, then $c^\varepsilon > 0$ and, at every point of the closed physical segment,*

$$|f^\varepsilon(u,\theta) - c^\varepsilon(\theta)| \le \mathcal{V}^\varepsilon(\theta) < \gamma^{-1}c^\varepsilon(\theta). \tag{8.1}$$

In particular $f^\varepsilon > 0$ and $W^\varepsilon > 0$ throughout that segment, including its one-sided endpoints. For each u at least one of the original tests satisfies

$$c_j(u) < 2\varepsilon q_0^{d_j/4} \quad \text{for some } 0 \le j \le m, \qquad \text{or} \qquad \Delta_j(u) < 2\varepsilon q_0^{d'_j/4} \quad \text{for some } 0 \le j < m. \tag{8.2}$$

The witnessing test may change with u.

Conversely, if W^ε vanishes anywhere on the closed segment, its complete original paired coefficient is paid: $c^{\varepsilon,\gamma}_{\mathrm{var}} = c^\varepsilon$ and $c^{\varepsilon,\gamma}_{\mathrm{per}} = 0$.

Proof. The endpoint minimum is attained at at least one endpoint, where $f^\varepsilon = c^\varepsilon$. Every increment from that endpoint is bounded by the total variation. Positivity of the remaining coefficient is precisely $c^\varepsilon > \gamma\mathcal{V}^\varepsilon$, which proves (8.1). If any visited interior incidence vanished at an endpoint, Theorem 4.3 would give $c^\varepsilon = 0$. Thus the segment belongs to $\mathcal{N}$. Its optical factor A is finite and positive on the closed segment: the exact formula $\omega = Rc_0c_m/\mathfrak{b}_m$ in (3.7) has a finite positive denominator when all visited incidences are positive, while (4.4) gives $\chi \ge \lambda_\circ$. This also applies at a critical endpoint with its fixed Gauss direction. Since $f^\varepsilon = AW^\varepsilon > 0$, the guard is positive there. If all inequalities in (8.2) failed, every factor defining H^ε would equal one, contradicting $W^\varepsilon = 1 - H^\varepsilon > 0$.

For the converse, let u_0 be a point where $W^\varepsilon = 0$. The finite optical factor gives $f^\varepsilon(u_0) = 0$, so the increment from an endpoint attaining c^ε gives $\mathcal{V}^\varepsilon \ge c^\varepsilon$. Apply (7.3). This argument also covers $u_0 = a$ or $u_0 = b$, with the proved physical one-sided values. □

The surviving class

$$\mathcal{P}_B = \{\theta \in \mathcal{N} : c^{\varepsilon(B)}(\theta) > \gamma_B\mathcal{V}^{\varepsilon(B)}(\theta)\} \tag{8.3}$$

is Borel. The source $\mathsf{U}^{\mathrm{per}}_B$ is supported on it; its coefficient is still the positive difference in (7.3), not the full original coefficient. The sum (7.8) retains every original coefficient exactly. No assumption that one fixed bad test covers the entire segment is made. A Borel witness at a specified point can be chosen as the first successful test in the finite ordered list in (8.2); this selection is not needed to prove the payment.

Remark 8.2 (Boundary mechanisms and what is retained). An unvisited third-disk tangency may give $W^{\varepsilon} = 1$ at an endpoint without forcing variation relative to the endpoint minimum. It is therefore not declared paid merely on account of its name. The same applies to a nongrazing critical entrance, an original section cut and a short-reach segment. Each receives precisely the two coefficients in (7.3). When the guard vanishes elsewhere on that segment the entire coefficient is paid by Theorem 8.1; otherwise a positive remaining coefficient is possible. At fixed collision count the optical data on a surviving segment have positive minima, but no such minimum is asserted uniformly in the collision count or the sheet.

8.1. **Comparison with the original nongrazing shortage.** Keep the *full original* nongrazing pool $P^{\mathrm{ng}}_{r_B}$, including its capacity on segments whose source has already been paid. Define only a new source-side remainder:

$$\mathsf{R}^{\mathrm{per}}_B(t) = \big(\mathsf{U}^{\mathrm{per}}_B(t) - K_B P^{\mathrm{ng}}_{r_B}(t)\big)_+. \tag{8.4}$$

The allocation by this pool is scalar at the roof, just as in Section 6. It is not a coupling of individual persistent sheets to individual comparison sheets.

Theorem 8.3 (Positive-source reduction with a paid error)**.** *On the original exact event, at every finite collision count,*

$$\mathsf{U}^{\mathrm{micro}}_B = \mathsf{U}^{\mathrm{var}}_B + \min\{\mathsf{U}^{\mathrm{per}}_B, K_B P^{\mathrm{ng}}_{r_B}\} + \mathsf{R}^{\mathrm{per}}_B, \tag{8.5}$$

$$0 \le \mathsf{R}^{\mathrm{ng}}_B - \mathsf{R}^{\mathrm{per}}_B \le \mathsf{U}^{\mathrm{var}}_B. \tag{8.6}$$

For every fixed M, in the prescribed order of limits,

$$\mathcal{H}_M(\mathsf{R}^{\mathrm{ng}}_B - \mathsf{R}^{\mathrm{per}}_B) \le C_M B^{-95/384}, \tag{8.7}$$

$$\mathcal{E}_M \le \mathcal{H}_M(\mathsf{R}^{\mathrm{per}}_B) + C_M B^{-1/384}. \tag{8.8}$$

The unchanged pointwise assertion (1.5) *is equivalent to $\mathcal{H}_M(\mathsf{R}^{\mathrm{per}}_B) \to 0$ for every fixed M.*

Proof. Equation (8.5) is the positive scalar identity $x = \min(x, y) + (x - y)_+$ applied to $x = \mathsf{U}^{\mathrm{per}}_B$, combined with (7.8). Monotonicity and the unit Lipschitz bound of the positive-part function give (8.6). Corollary 7.3 then proves (8.7).

For completeness, the height of the middle term in (8.5) is at most

$$\mathcal{H}_M(K_B P_{r_B}) \le C\varepsilon^{-1/32}\{\varepsilon^{1/16} + (1 + \varepsilon^{-1})\Delta_B\} = O(\varepsilon^{1/32}) = O(B^{-1/384}),$$

by Theorem 5.1. Insert the three nonnegative terms in Input statement 2.2, and use subadditivity of the ordered upper height. This proves (8.8) and sufficiency of the displayed limit. Necessity follows from the same inherited scalar equivalence and $0 \le \mathsf{R}^{\mathrm{per}}_B \le \mathsf{U}^{\mathrm{micro}}_B$. No essential supremum is interchanged with an integral, and the comparison pool has not been restricted to $\mathcal{P}_B$. □

The estimate still required to conclude (1.5) is

$$\lim_{B\to\infty} \limsup_{m\to\infty} \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} \big(\mathsf{U}^{\mathrm{per}}_{B,m}(t) - K_B P^{\mathrm{ng}}_{r_B,m}(t)\big)_+ = 0. \tag{8.9}$$

Theorems 7.2 and 8.1 prove an actual summed payment and a necessary physical persistence property. They do not prove (8.9). In particular a positive nearly constant density on a shrinking segment need not have small pointwise height merely because its roof integral is small. The original raw pointwise theorem, and not an averaged substitute, remains the organizing target.

9. Global uniqueness on a roof–angle fiber

The estimates in this section concern actual finite trajectories, not a continuation through a forbidden event. In particular the regular domain of a lifted itinerary need not be connected or convex. This distinction is important when the original section and occupation tests divide that domain into many components.

Normalize the initial lifted scatterer to $O_0 = 0$. Let $\mathcal{W}_m(R)$ denote the realizable lifted itineraries of m flights. This set is finite by the finite-horizon bound and the lattice geometry. Fix $w = (O_0, \dots, O_m) \in \mathcal{W}_m(R)$. For a physical trajectory write $Q_j = O_j + Rn_j$, let t_j be the counterclockwise rotation of n_j, and put

$$S(Q) = \sum_{j=0}^{m-1} |Q_{j+1} - Q_j|, \qquad g = R(-p_0, p_m), \qquad z = |g|.$$

As a function of the independent Cartesian contacts, S is convex. It is differentiable at every configuration in question because successive scatterer disks are disjoint. Write $\beta_0 = \beta_m = 1$ and $\beta_j = 2$ for $0 < j < m$.

Lemma 9.1 (Two-trajectory support inequalities)**.** *Let Q and $\widehat{Q}$ be two regular physical trajectories with the same lifted itinerary and radius. Set*

$$b_j = 1 - n_j \cdot \widehat{n}_j \geq 0, \qquad s_j = t_j \cdot \widehat{n}_j, \qquad s_E = (s_0, s_m).$$

Their lengths satisfy

$$S(\widehat{Q}) - S(Q) \geq R \sum_{j=0}^{m} \beta_j c_j b_j + g \cdot s_E, \tag{9.1}$$

$$S(Q) - S(\widehat{Q}) \geq R \sum_{j=0}^{m} \beta_j \widehat{c}_j b_j - \widehat{g} \cdot s_E. \tag{9.2}$$

Consequently

$$(\widehat{g} - g) \cdot s_E \geq R \sum_{j=0}^{m} \beta_j (c_j + \widehat{c}_j) b_j. \tag{9.3}$$

No interpolation by physical trajectories is used.

Proof. The physical reflection law and first variation in Cartesian coordinates give

$$\nabla_0 S = -c_0 n_0 - p_0 t_0, \qquad \nabla_j S = -2c_j n_j \ (0 < j < m), \qquad \nabla_m S = -c_m n_m + p_m t_m.$$

The convex support inequality is $S(\widehat{Q}) - S(Q) \geq \sum_j \nabla_j S(Q) \cdot (\widehat{Q}_j - Q_j)$. Since $\widehat{Q}_j - Q_j = R(\widehat{n}_j - n_j)$, its normal terms are $R\beta_j c_j b_j$ and its two tangential terms are $g \cdot s_E$. This proves (9.1). Reversing the roles of the two configurations leaves b_j unchanged and replaces s_j by $-s_j$. This proves (9.2). Adding the two inequalities proves (9.3). The support inequality is applied to a convex Cartesian function; it requires neither a convex angular chart nor a physical path joining the configurations. □

Theorem 9.2 (Global roof–angle uniqueness)**.** *For a fixed lifted itinerary, radius and direction $e \in S^1$, the physical length is strictly increasing among regular trajectories whose endpoint gradients are $g = ze$, $z > 0$, ordered by z. At a fixed z there is at most one such trajectory. In particular*

$$Q \longmapsto (F(Q), \arg g(Q)) \tag{9.4}$$

is globally injective on the noncritical regular trajectories of that itinerary. This remains true after any of the original exact event tests, and across all their disconnected regular components.

Proof. If $\widehat{g} = g$, the left side of (9.3) is zero. All c_j and $\widehat{c}_j$ are strictly positive, so every $b_j = 0$. Thus all physical contacts, and then all flight directions, coincide.

Now let $\widehat{g} = \widehat{z}e$, $g = ze$ and $\widehat{z} > z > 0$. The configurations cannot coincide. At least one b_j is therefore positive, and (9.3) gives

$$(\widehat{z} - z)e \cdot s_E \geq R \sum_j \beta_j (c_j + \widehat{c}_j) b_j > 0.$$

Hence $e \cdot s_E > 0$. The right side of (9.1) is then strictly positive, proving $S(\widehat{Q}) > S(Q)$. Equal roof and angle would force equal z, and the first part of the proof would force equality of trajectories. Restricting an injective map to the original exact event cannot increase its multiplicity. □

Lemma 9.3 (A bounded roof range for each physical itinerary)**.** *For any two regular physical trajectories with the same itinerary,*

$$|S(Q) - S(\widehat{Q})| \leq |Q_0 - \widehat{Q}_0| + |Q_m - \widehat{Q}_m| \leq 4R. \tag{9.5}$$

Proof. With its endpoint contacts fixed, a regular physical trajectory minimizes S over the product of the closed interior scatterer disks. Indeed, for a competitor Y_j in the jth disk, $n_j \cdot (Y_j - Q_j) \leq 0$; the convex support inequality and $\nabla_j S = -2c_j n_j$ therefore give the minimizing property. Use the interior contacts of $\widehat{Q}$ as a competitor for the endpoints of Q. Changing only the first and last endpoint changes the sum of flight lengths by at most the sum of their displacements, by the triangle inequality. This proves one direction of (9.5); interchanging the trajectories proves the other. The same triangle inequality covers $m = 1$. □

This range estimate does not control how many original occupation labels occur in a word domain. In particular it is not used to replace the exact occupation test by an unqualified interval test.

9.1. **The original physical sum at a target roof.** For the original section tests, $\chi \geq \lambda_\circ$ by (4.4). On each regular component, the map (9.4) is a local diffeomorphism. Theorem 9.2 shows that all those components of a fixed itinerary have disjoint images, not merely that each component has a nonsingular Jacobian. The equal-gradient argument also gives at most one regular critical trajectory per word, even when the gradient is zero. There are finitely many physical words at each finite count, so these critical points have zero collision area. Consequently, for any nonnegative Borel trajectory weight a, the original roof density has the representation

$$\frac{dF_{\#}(a\mathbf{1}_\Pi \nu_R^*)}{dt}(t) = \int_0^{2\pi} \sum_{w \in \mathcal{W}_m(R)} \mathbf{1}_{\{x_w(t,\vartheta) \text{ exists}\}} \frac{\omega(x_w) a(x_w)}{4\pi c_* \chi(x_w)} \, d\vartheta. \tag{9.6}$$

Here x_w is the unique point of the original exact event with the indicated word, roof and angle. A missing point contributes zero. The representation holds for almost every roof, in the original order $\sup_R \operatorname{ess\,sup}_t$. Angular chart overlap is removed before summation; the incident-component convention remains unchanged.

Proposition 9.4 (Monotone completion of the target-roof formula)**.** *Formula* (9.6) *is an identity of nonnegative measures for every nonnegative Borel weight, with extended values allowed before an integrability bound is proved. In particular it applies to the unbounded products* $\prod_{i \in S} c_i^{-1}$.

Proof. Fix m, R, n, k. Cover the noncritical regular physical domain in endpoint coordinates by countably many relatively compact local charts V_j on which $(F, \arg g)$ is a C^1 diffeomorphism into an unwrapped angular chart. Such a cover exists by the nonvanishing Jacobian and second countability. Subdivide as necessary so that both maps have bounded derivatives on each smaller chart. Disjointize the source charts by $B_j = V_j \setminus \bigcup_{k<j} V_k$. The sets B_j are Borel; no chart overlap is counted twice. Angular representatives are also split at a fixed seam, so their range is $[0, 2\pi)$.

For a nonnegative Borel weight h put

$$h_N = \min\{h, N\}\mathbf{1}_{\bigcup_{j\le N} B_j}.$$

This sequence increases to h on the regular noncritical domain. Apply ordinary change of variables on each of the first N charts, restricted to its Borel piece, and test against a nonnegative Borel function of (t, ϑ). The local inverse Jacobian, multiplied by the original collision density, is exactly $\omega/(4\pi c_*\chi)$. Local inverses are continuous and agree where they represent the same physical point, so their Borel restrictions define the displayed x_w without a selection ambiguity. Global uniqueness prevents different regular components of one word from producing the same target. Distinct words are summed, not identified.

Monotone convergence on the source, followed by Tonelli and monotone convergence on the target, now gives the identity for h as an extended nonnegative measure. Only after this passage is an upper pressure bound used to establish finiteness and an L^∞ density. This avoids assuming the integrability of the unbounded marked weight in order to prove its own change of variables.

The complement of the two-sided regular set has zero collision area. On the countably many charts just chosen, absolute continuity of the coordinate changes and their local Lipschitz property give zero target area for its image. The finitely many regular critical points also have zero source area. Removing these sets therefore does not change the weighted source measure. Fubini yields the claimed roof identity for almost every t. All null sets here are chosen at the fixed radius; no exceptional set common to every R is asserted or needed. For the finitely many marked sets at a fixed m, intersecting the corresponding full-measure roof sets provides a common version for their finite positive expansions. □

10. A complete marked pressure bound at the target roof

We now transfer the retained geometric pressure estimate to the roof fibers in (9.6). The previous momentum-fiber bound alone would not justify this transfer: different words at the same roof can have different endpoint momentum magnitudes. The global injectivity just proved supplies the missing multiplicity statement.

Let $j(x) = J^s T_R(x)$ on the full-area set M'_R regular in both infinite time directions, and let $j_l(x) = \prod_{v=0}^{l-1} j(T_R^v x)$, with $j_0 = 1$. The lifted physical one-flight alphabet has at most D members, uniformly in R. Define

$$P_l(R) = \sum_{w\in\mathcal{W}_l(R)} \sup_{x\in[w]_R\cap M'_R} j_l(x), \qquad P_0 = 1.$$

Input statement 10.1 (Uniform physical pressure and stable optical comparison)**.** On the circular family $R \in [9/20, 47/100]$, use the original finite-horizon flight bounds, positive curvature bounds and the original endpoint section cap. The stable Jacobian j is computed with Euclidean arclength along the true stable line in collision coordinates (s, φ), where $s = R\alpha$ and $p = \sin\varphi$. It is restricted to M'_R, the full-area set regular in both infinite directions. Its cocycle uses this same convention on every block, including the empty block. The endpoint weight ω is still the derivative in the original (α, p) coordinates of (2.1); all coordinate factors are included in the uniform optical constant.

For the realizable lifted physical words just defined, the retained geometric theorems state

$$\sup_{R,l} P_l(R) \le Q_*, \qquad \omega(x) \le C_0 c_m(x) j_m(x), \qquad j(x)/c(x) \le C_1. \tag{10.1}$$

All three constants are independent of R, l and the original exact labels. These statements concern positive, unconditioned physical word sums; they are not a twisted or exact-label pressure theorem.

We take $Q_* \geq 1$ and $C_0, C_1 > 0$. These are the uniform geometric pressure theorem in historical module 241 and the stable optical comparison in module 238, with the physical-word domination in module 239. Their complete original sources are pinned in the proof ledger and copied in the provenance directory. They involve no occupation, displacement or roof test. Such tests are dropped only on the upper side of the estimates below. No new pressure estimate for a twisted or point-conditioned operator is assumed.

Lemma 10.2 (Removal of several marked stable factors)**.** *For $S = \{i_1 < \cdots < i_s\} \subset \{1, \dots, m-1\}$, at every regular trajectory in M'_R,*

$$\frac{\omega(x)}{\prod_{i\in S} c_i(x)} \leq C_0 C_1^s \prod_{v\in\{0,\dots,m-1\}\setminus S} j(T_R^v x). \tag{10.2}$$

Consequently, if at most one point x_w is chosen from each physical word domain, then

$$\sum_w \frac{\omega(x_w)}{\prod_{i\in S} c_i(x_w)} \leq C_0 C_1^s D^s Q_*^{s+1}. \tag{10.3}$$

The choice of point may depend on the word and need not have the same momentum target for different words.

Proof. Use $c_m \leq 1$ in (10.1), factor j_m into its individual steps, and apply $j(T_R^i x)/c_i(x) \leq C_1$ at each marked index. This gives (10.2).

Delete the s marked steps of a word. The remaining blocks have lengths

$$i_1, \quad i_2 - i_1 - 1, \ \dots, \ i_s - i_{s-1} - 1, \quad m - i_s - 1.$$

For clarity, steps are numbered $0, \dots, m-1$, whereas the marks belong to $1, \dots, m-1$. Their indexing is

$$\underbrace{0, \dots, i_1 - 1}_{i_1} \mid i_1 \mid \underbrace{i_1 + 1, \dots, i_2 - 1}_{i_2 - i_1 - 1} \mid i_2 \mid \cdots \mid i_s \mid \underbrace{i_s + 1, \dots, m-1}_{m - i_s - 1}.$$

For example, $m = 7$ and $S = \{1, 2, 6\}$ leave blocks of lengths $(1, 0, 3, 0)$ and delete three physical symbols. Thus adjacent marks and a final mark are included, and every empty block contributes $P_0 = 1$. Some lengths can be zero. The blocks, translated to the initial fundamental scatterer, together with the s deleted physical step symbols determine the full lifted word uniquely. There are at most D^s choices for those symbols. Bound the remaining cocycle on each block by its own supremum and sum over all block words. Positivity permits adding combinations which do not produce a realizable full word. The resulting upper bound is $C_0 C_1^s D^s$ times the product of the $s+1$ corresponding $P_l(R)$'s, at most $C_0 C_1^s D^s Q_*^{s+1}$. For $S = \varnothing$ it is simply $C_0 P_m(R)$. This argument counts physical step symbols, never the infinitely many homogeneous strips at a removed incidence. □

Set

$$C_\star = \frac{C_0 Q_*}{2 c_* \lambda_\circ}, \qquad K_\star = C_1 D Q_*. \tag{10.4}$$

Neither constant depends on the collision count, radius, exact labels, roof or marked indices.

Theorem 10.3 (Target-roof multiple-incidence insertion)**.** *For every finite m, every original exact event and every $S \subset \{1, \dots, m-1\}$, the nonnegative measure*

$$F_\# \left(\prod_{i\in S} c_i^{-1} \mathbf{1}_\Pi \nu_R^* \right)$$

has a roof density d_S satisfying

$$\|d_S\|_{L^\infty(dt)} \leq C_\star K_\star^{|S|}. \tag{10.5}$$

The bound holds on the whole roof line. For arbitrary numbers $a_1, \dots, a_{m-1} \geq 0$, the density $d_{\boldsymbol{a}}$ with weight $\prod_{i=1}^{m-1}(1 + a_i/c_i)$ satisfies

$$\|d_{\boldsymbol{a}}\|_\infty \leq C_\star \prod_{i=1}^{m-1} (1 + K_\star a_i) \leq C_\star \exp\left(K_\star \sum_{i=1}^{m-1} a_i \right). \tag{10.6}$$

In particular the complete inherited linear incidence weight obeys

$$\left\| \frac{dF_\#(I_m \mathbf{1}_\Pi \nu_R^*)}{dt} \right\|_\infty \leq C_\star \left(1 + K_\star \frac{2q_0}{1 - q_0} \right). \tag{10.7}$$

These are finite-count unscaled height bounds, not m^{-2} local limit estimates.

Proof. At an almost-everywhere target (t, ϑ), Theorem 9.2 supplies at most one x_w per word. Apply (10.3) to precisely those points, use $\chi \geq \lambda_\circ$, and integrate (9.6) over the angle interval of length 2π. This gives (10.5) with the constants (10.4). Proposition 9.4 first establishes the nonnegative change-of-variables identity by truncation and monotone exhaustion; the present bound then proves finiteness and the density. At finite m the roof is confined to a bounded interval.

Expand $\prod_i (1 + a_i/c_i)$ into its finite sum over distinct marked sets. Apply (10.5) to each term. The sum of the upper bounds factors exactly as $C_\star \prod_i (1 + K_\star a_i)$. The exponential inequality uses $1 + x \leq e^x$. Finally expand I_m linearly and use $\sum_i q_0^{d_i} \leq 2q_0/(1 - q_0)$. All original exact labels remain in the source throughout. Removing their tests only enlarges the positive word upper sums in the proof; it does not change the source normalization or infer a Gaussian factor from an unconditioned pressure bound. □

Corollary 10.4 (A factorial bound for changing incidence witnesses)**.** *Let $J \subset \{1, \dots, m-1\}$ and let $\eta_i \geq 0$ be arbitrary. On the original event define*

$$N_J(x) = \sum_{i \in J} \mathbf{1}_{\{c_i(x) < \eta_i\}}, \qquad A_J = \sum_{i \in J} \eta_i.$$

For every integer $s \geq 1$,

$$\left\| \frac{dF_\#(\mathbf{1}_{\{N_J \geq s\}} \mathbf{1}_\Pi \nu_R^*)}{dt} \right\|_\infty \leq C_\star K_\star^s e_s((\eta_i)_{i \in J}) \leq C_\star \frac{(K_\star A_J)^s}{s!}, \tag{10.8}$$

where $e_s = 0$ for $s > |J|$. The witnesses may change with the target roof and angle; no single witness is chosen for an entire segment.

Proof. At every regular state,

$$\mathbf{1}_{\{N_J \geq s\}} \leq \binom{N_J}{s} = \sum_{\substack{S \subset J \\ |S| = s}} \prod_{i \in S} \mathbf{1}_{\{c_i < \eta_i\}} \leq \sum_{\substack{S \subset J \\ |S| = s}} \prod_{i \in S} \eta_i / c_i.$$

Push this positive inequality through the original roof and apply (10.5). In the expansion of A_J^s each product with distinct indices occurs $s!$ times; all remaining terms are nonnegative. This proves the last inequality. The argument is finite-count and already completes the full physical sum, so it introduces no collision-count multiplier for the possible witnesses. □

11. Pointwise removal of deep and numerous incidence witnesses

We apply the marked pressure theorem to the unchanged positive persistent source of Section 8. The new payment concerns its value at the original target roof, rather than an average in that roof. Clearance witnesses require a separate estimate and will not be identified with incidence witnesses.

Lemma 11.1 (Domination of the surviving coefficient at every roof)**.** *For $\gamma \geq 1$, on every original physical segment and every point of its closed interval,*

$$0 \leq c_{\mathrm{per}}^{\varepsilon,\gamma} \leq f^{\varepsilon}(u,\theta) \leq A(u,\theta). \tag{11.1}$$

Proof. Total variation bounds the decrement from an endpoint attaining c^{ε}, so $f^{\varepsilon}(u) \geq c^{\varepsilon} - \mathcal{V}^{\varepsilon}$. If $c_{\mathrm{per}} > 0$, its value is $c^{\varepsilon} - \gamma\mathcal{V}^{\varepsilon} \leq c^{\varepsilon} - \mathcal{V}^{\varepsilon}$. If $c_{\mathrm{per}} = 0$, nonnegativity suffices. The last inequality is the unchanged bound $0 \leq W^{\varepsilon} \leq 1$. □

Thus a target-dependent Borel restriction of the persistent paired source is bounded by the corresponding restriction of the original uncut roof measure. This assertion concerns the surviving coefficient, not the full endpoint minimum on an arbitrary segment.

For $\varepsilon > 0$, use precisely the incidence witness thresholds already forced by the original guard:

$$E_i^{\varepsilon} = \{c_i < \eta_i^{\varepsilon}\}, \qquad \eta_i^{\varepsilon} = 2\varepsilon q_0^{d_i/4}, \quad d_i = \min(i, m-i), \quad 1 \leq i < m. \tag{11.2}$$

Write $N^{\varepsilon} = \sum_{i=1}^{m-1} \mathbf{1}_{E_i^{\varepsilon}}$. There are at most two indices at each depth. Hence, for an integer $L \geq 1$,

$$\sum_{d_i \geq L} \eta_i^{\varepsilon} \leq \frac{4\varepsilon q_0^{L/4}}{1 - q_0^{1/4}}, \qquad \sum_{i=1}^{m-1} \eta_i^{\varepsilon} \leq A_{\varepsilon} := \frac{4\varepsilon q_0^{1/4}}{1 - q_0^{1/4}}. \tag{11.3}$$

For any Borel target event E, let

$$\mathsf{U}_B^{\mathrm{per}}[E](t) = \int c_{\mathrm{per}}^{\varepsilon(B),\gamma_B}(\theta)\mathbf{1}_{\{a<t<b\}}\widetilde{H}_B(\Psi(\theta,t))\mathbf{1}_E(\Psi(\theta,t))\,d\theta.$$

These are restrictions of the source in (7.8); no new normalization is used.

Theorem 11.2 (Uniform target-roof witness estimates)**.** *For every finite m, every B and integers $L, s \geq 1$,*

$$\|\mathsf{U}_B^{\mathrm{per}}[\bigcup_{d_i \geq L} E_i^{\varepsilon(B)}]\|_{\infty} \leq \frac{4C_\star K_\star \varepsilon(B)}{1 - q_0^{1/4}} q_0^{L/4}, \tag{11.4}$$

$$\|\mathsf{U}_B^{\mathrm{per}}[N^{\varepsilon(B)} \geq s]\|_{\infty} \leq C_\star \frac{(K_\star A_{\varepsilon(B)})^s}{s!}. \tag{11.5}$$

The norms are essential suprema on the whole roof line; constants are uniform in the radius and original exact labels.

Fix any $\eta > 0$ and define, at finite count,

$$\begin{aligned} L_m &= \left\lceil \frac{4(2+\eta)}{|\log q_0|} \log(m+1) \right\rceil, \\ s_m &= \left\lceil \frac{4\log(m+e)}{\log\log(m+e^e)} \right\rceil, \\ \mathcal{D}_{B,m} &= \bigcup_{d_i \geq L_m} E_i^{\varepsilon(B)}, \qquad \mathcal{A}_{B,m} = \{N^{\varepsilon(B)} \geq s_m\}, \qquad \mathcal{Z}_{B,m} = \mathcal{D}_{B,m} \cup \mathcal{A}_{B,m}. \end{aligned} \tag{11.6}$$

An empty union contributes zero. For every fixed outer B and M,

$$\mathcal{H}_M(\mathsf{U}_B^{\mathrm{per}}[\mathcal{Z}_{B,m}]) = 0. \tag{11.7}$$

More precisely the deep part satisfies, at every count,

$$\sup_{R,n,k} m^2 \|\mathsf{U}_B^{\mathrm{per}}[\mathcal{D}_{B,m}]\|_{\infty} \leq C\varepsilon(B)(m+1)^{-\eta}. \tag{11.8}$$

Proof. By Lemma 11.1, each source in the first two assertions is bounded by the original uncut density restricted to the same target event. Apply Corollary 10.4 with $s = 1$ to the depth tail and with general s to all indices, then use (11.3). This proves the two finite-count bounds after the physical sum, with no selected uniform witness.

They hold for every L and s at every m. Substituting L_m in the first gives $q_0^{L_m/4} \le (m{+}1)^{-(2+\eta)}$, which proves (11.8). For fixed B, put $a = K_\star A_{\varepsilon(B)} < \infty$. The elementary factorial bound $s! \ge (s/e)^s$ gives

$$m^2 \frac{a^{s_m}}{s_m!} \le \exp\{2\log m + s_m(1 + \log a - \log s_m)\} = m^{-2+o(1)} \longrightarrow 0.$$

Here $s_m \log s_m = (4{+}o(1)) \log m$ and $s_m = o(\log m)$. The case $a = 0$ is immediate. If $s_m > m-1$, the event is empty. Subadditivity and positivity complete the proof of (11.7).

The depth and multiplicity thresholds in this proof are substitutions into finite-count inequalities valid simultaneously for all their values. No fixed-window spectral estimate is evaluated at a collision-dependent window. The original outer quantities $\varepsilon(B), h_B, r_B, \gamma_B, K_B$ remain fixed before the collision limit. Thus the proof does not exchange that limit with any outer width or band. □

11.1. **The retained source after this pointwise payment.** Define

$$\begin{aligned} \mathsf{U}_B^{\mathrm{loc}} &= \mathsf{U}_B^{\mathrm{per}}[\mathcal{Z}_{B,m}^{c}], \\ \mathsf{U}_B^{\mathrm{exc}} &= \mathsf{U}_B^{\mathrm{per}}[\mathcal{Z}_{B,m}], \qquad \mathsf{R}_B^{\mathrm{loc}} = (\mathsf{U}_B^{\mathrm{loc}} - K_B P_{r_B}^{\mathrm{ng}})_+. \end{aligned} \tag{11.9}$$

The superscript loc records localization of all incidence witnesses to the two collision-end layers; it makes no claim that clearance witnesses are localized to those layers. The comparison pool is the same full nongrazing pool, without a source-dependent restriction.

Corollary 11.3 (Exact preservation and collision-zero residual difference)**.** *At finite count the original persistent source satisfies*

$$\mathsf{U}_B^{\mathrm{per}} = \mathsf{U}_B^{\mathrm{exc}} + \mathsf{U}_B^{\mathrm{loc}}, \qquad 0 \le \mathsf{R}_B^{\mathrm{per}} - \mathsf{R}_B^{\mathrm{loc}} \le \mathsf{U}_B^{\mathrm{exc}}. \tag{11.10}$$

For every fixed B, M,

$$\mathcal{H}_M(\mathsf{R}_B^{\mathrm{per}} - \mathsf{R}_B^{\mathrm{loc}}) = 0, \qquad \mathcal{E}_M \le \mathcal{H}_M(\mathsf{R}_B^{\mathrm{loc}}) + C_M B^{-1/384}. \tag{11.11}$$

The unchanged raw pointwise assertion is equivalent to $\mathcal{H}_M(\mathsf{R}_B^{\mathrm{loc}}) \to 0$ for every fixed M.

Proof. The source partition is an identity of nonnegative original submeasures. Monotonicity and the unit Lipschitz bound of the positive-part map give its residual inequality. Apply (11.7) and then Theorem 8.3. For necessity use $0 \le \mathsf{R}_B^{\mathrm{loc}} \le \mathsf{U}_B^{\mathrm{micro}}$ and the retained scalar equivalence. Every removed term has been retained in the identity and paid in the target essential-height topology. □

For all sufficiently large B, $2\varepsilon(B) < c_\circ$, so the original endpoint section cap rules out incidence witnesses at $i = 0, m$. At a target point with positive surviving coefficient the original guard still requires an incidence or a clearance witness, by Theorem 8.1. On the retained part, every incidence witness has depth below L_m, and fewer than s_m such witnesses occur. A clearance witness may occur at any original flight, with its original positive weight. These alternatives allow a witness which changes with the roof; they assert no common witness on an entire physical segment.

The precise remaining estimate is

$$\lim_{B\to\infty} \limsup_{m\to\infty} \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} (\mathsf{U}_{B,m}^{\mathrm{loc}}(t) - K_B P_{r_B,m}^{\mathrm{ng}}(t))_+ = 0. \tag{11.12}$$

The target-roof pressure theorem proves a new physical weighted summation and the pointwise removal in Theorem 11.2; it does not prove (11.12). In particular the all-incidence bound $C\varepsilon$ furnished by (11.4) with $L = 1$ is unscaled. At a fixed outer B it cannot be multiplied by m^2 and declared small. Nor does an inverse-incidence insertion estimate an inverse third-disk clearance. The remaining source includes the original shallow incidence witnesses, unvisited tangencies, critical entrances, section cuts and capacity deficiencies which are not paid by the proved estimates. No change of the original theorem or of its arithmetic classes is made in stating this boundary.

12. Joint depth and severity of incidence witnesses

The factorial estimate treats the witness thresholds only through their sum. It therefore forgets that there are at most two collision indices at any depth. It also forgets how much an individual incidence lies below its guard threshold. We retain both pieces of information. All estimates below concern the same target-roof measure as Theorem 10.3, with its original exact labels. They use its inherited geometric inputs, not a new conditional mixing or local-limit assumption.

Put

$$r = q_0^{1/4}, \qquad \alpha = -\log r = \tfrac{1}{4}|\log q_0|, \qquad b_\varepsilon = 4K_\star\varepsilon/\alpha. \tag{12.1}$$

For a regular trajectory define its incidence-witness set and load by

$$\begin{aligned} J_\varepsilon(x) &= \{i : 1 \le i < m,\ c_i(x) < 2\varepsilon r^{d_i}\},\\ \mathscr{L}_\varepsilon(x) &= \sum_{i\in J_\varepsilon(x)} \log\frac{2\varepsilon}{c_i(x)}\\ &= \alpha \sum_{i\in J_\varepsilon(x)} d_i + \sum_{i\in J_\varepsilon(x)} \log\frac{2\varepsilon r^{d_i}}{c_i(x)}. \end{aligned} \tag{12.2}$$

The empty sum is zero. Both summands in the last line are nonnegative. In particular this is not a signed logarithmic weight, and it introduces no clearance factor. The Borel witness set is allowed to vary with the roof and angle.

For a nonnegative Borel weight h, write

$$D[h](t) = \frac{dF_\#(h\mathbf{1}_\Pi \nu_R^*)}{dt}(t)$$

when this density is defined. The positive identities of the preceding section give a common version at fixed m, R, n, k. The next theorem also proves existence and finiteness for its weights.

Theorem 12.1 (Exponential moment of the original incidence load)**.** *For every finite collision count, every $\varepsilon > 0$ and $0 < \lambda < 1$,*

$$\left\|D[e^{\lambda\mathscr{L}_\varepsilon}]\right\|_\infty \le C_\star \exp\left\{\frac{4K_\star\varepsilon}{e^{\alpha(1-\lambda)}-1}\right\}. \tag{12.3}$$

Consequently, for $u > b_\varepsilon$,

$$\left\|D[\mathbf{1}_{\{\mathscr{L}_\varepsilon \ge u\}}]\right\|_\infty \le C_\star \min\left\{1, \exp[-u + 2\sqrt{b_\varepsilon u}]\right\}. \tag{12.4}$$

Both estimates are uniform in the radius, exact labels and target roof. They remain valid with any further Borel restriction of the original measure, including the persistent source through Lemma 11.1.

Proof. Set $y_i = 2\varepsilon/c_i$. At a witness, $y_i > r^{-d_i}$, whence

$$y_i^\lambda = y_i y_i^{-(1-\lambda)} \le y_i r^{(1-\lambda)d_i}.$$

At every regular trajectory we therefore have the positive inequality

$$e^{\lambda \mathscr{L}_\varepsilon} = \prod_{i=1}^{m-1} \left[1 + (y_i^\lambda - 1)\mathbf{1}_{\{i \in J_\varepsilon\}}\right] \le \prod_{i=1}^{m-1} \left(1 + \frac{2\varepsilon r^{(1-\lambda)d_i}}{c_i}\right).$$

Apply the finite product insertion (10.6). There are at most two indices at every positive depth, so

$$\sum_{i=1}^{m-1} 2\varepsilon r^{(1-\lambda)d_i} \le 4\varepsilon \sum_{d \ge 1} r^{(1-\lambda)d} = \frac{4\varepsilon}{e^{\alpha(1-\lambda)} - 1}.$$

This proves (12.3) after the complete physical sum. Every product is finite before this last geometric upper bound is used; no independence of collisions is assumed.

Positivity gives $\mathbf{1}_{\{\mathscr{L}_\varepsilon \ge u\}} \le e^{-\lambda u} e^{\lambda \mathscr{L}_\varepsilon}$. Writing $v = 1 - \lambda$ and using $e^{\alpha v} - 1 \ge \alpha v$, we obtain, for every $0 < v < 1$,

$$\left\| D[\mathbf{1}_{\{\mathscr{L}_\varepsilon \ge u\}}] \right\|_\infty \le C_\star \exp\{-u + vu + b_\varepsilon / v\}.$$

When $u > b_\varepsilon$, take $v = \sqrt{b_\varepsilon / u}$. The independent bound $\|D[1]\|_\infty \le C_\star$ follows from the empty marked set and supplies the minimum in (12.4). Borel restrictions decrease a positive measure. For the persistent source the pointwise coefficient inequality in Lemma 11.1 supplies this decrease. □

Corollary 12.2 (Accumulated depth and severe shallow incidences)**.** *For $v > 0$, each of the events*

$$\left\{ \alpha \sum_{i \in J_\varepsilon} d_i \ge v \right\}, \qquad \left\{ \sum_{i \in J_\varepsilon} \log \frac{2\varepsilon r^{d_i}}{c_i} \ge v \right\}$$

is contained in $\{\mathscr{L}_\varepsilon \ge v\}$ and has the bound (12.4) *whenever $v > b_\varepsilon$. An incidence witness satisfying $c_i \le 2\varepsilon e^{-v}$ is also in that event, irrespective of its depth.*

Proof. Use the nonnegative decomposition in (12.2). For the last assertion, the single indicated summand is at least v. □

Thus a single sufficiently severe witness in an end layer can be paid. The conclusion does not require either a deep witness or the large witness count of Theorem 11.2. It does not assert a bound for an unvisited third-disk tangency.

13. Quadratic tails for the number of incidence witnesses

There is also an explicit finite-count improvement which retains all threshold locations, rather than passing to their total sum. The calculation is deterministic; it makes no independence claim about the billiard or its symbolic itinerary.

For $0 < r < 1$ write

$$(r;r)_a = \prod_{j=1}^{a} (1 - r^j), \quad (r;r)_0 = 1, \quad (r;r)_\infty = \prod_{j=1}^{\infty} (1 - r^j) > 0, \quad f(s) = \left\lfloor \frac{(s+1)^2}{4} \right\rfloor.$$

The infinite product is positive because the tail of $\sum_j -\log(1 - r^j)$ is bounded by a constant times $\sum_j r^j$.

Lemma 13.1 (The two-end depth polynomial)**.** *For $0 \le s \le m-1$, put $E_{m,s}(r) = e_s((r^{d_i})_{1\le i<m})$. Then*

$$r^{f(s)} \le E_{m,s}(r) \le \sum_{a=0}^{s} \frac{r^{a(a+1)/2+(s-a)(s-a+1)/2}}{(r;r)_a (r;r)_{s-a}} \le \frac{(s+1)r^{f(s)}}{(r;r)_\infty^2}. \tag{13.1}$$

For $s > m-1$ the finite polynomial coefficient is zero and only the upper inequalities are asserted. In particular the coefficient has quadratic logarithmic rate, uniformly over $m \ge s+1$:

$$\log E_{m,s}(r) = -\frac{\alpha}{4}s^2 + O_r(s + \log(s+1)). \tag{13.2}$$

This is sharpness of the depth polynomial, not a lower bound for a physical roof density.

Proof. The multiset of finite collision depths is contained in two copies of $1, 2, 3, \ldots$. Coefficientwise positivity bounds its generating polynomial by $\prod_{d\ge1}(1+zr^d)^2$. To compute one factor, for $a \ge 1$ write $1 \le d_1 < \cdots < d_a$ as $d_j = j + h_j$, where $0 \le h_1 \le \cdots \le h_a$. Setting $g_1 = h_1$ and $g_j = h_j - h_{j-1}$ gives

$$\sum_{1\le d_1<\cdots<d_a} r^{d_1+\cdots+d_a} = r^{a(a+1)/2} \prod_{j=1}^{a} \sum_{g_j\ge0} r^{(a-j+1)g_j} = \frac{r^{a(a+1)/2}}{(r;r)_a}.$$

The empty factor contributes one. Multiplying the two positive series proves the middle sum in (13.1); all infinite passages are monotone limits of nonnegative coefficients. The minimum of its exponent over a is $f(s)$, attained by the two most balanced choices. Each denominator is at least $(r;r)_\infty^2$, giving the last inequality.

For the lower bound take the s smallest available depths. If $s = 2k$ these consist of two copies of $1, \ldots, k$ and have sum $k(k+1)$. If $s = 2k+1$ take these and one copy of $k+1$, with sum $(k+1)^2$. These depths are available whenever $m \ge s+1$, including the single central depth when m is even. Their product is one term of $E_{m,s}$. Taking logarithms of the two bounds proves (13.2). □

Theorem 13.2 (Quadratic witness-count tail at the target roof)**.** *With $N^\varepsilon = |J_\varepsilon|$, for every $s \ge 1$,*

$$\left\| D[\mathbf{1}_{\{N^\varepsilon \ge s\}}] \right\|_\infty \le \frac{C_\star (s+1)(2K_\star\varepsilon)^s}{(r;r)_\infty^2}\, r^{f(s)}. \tag{13.3}$$

The event is empty for $s > m-1$. The same estimate holds for the persistent source. For every fixed B and every $\eta > 0$, take

$$\sigma_m = \left\lceil 2\sqrt{\frac{(2+\eta)\log(m+1)}{\alpha}} \right\rceil. \tag{13.4}$$

Then the source with at least σ_m incidence witnesses has zero m^2-normalized essential height, with finite-count rate $m^{-\eta+o_B(1)}$.

Proof. Use the first, unrelaxed inequality in (10.8), with thresholds $2\varepsilon r^{d_i}$, and apply Lemma 13.1. Since $f(s) \ge s^2/4$, the logarithm of m^2 times the upper bound at $s = \sigma_m$ is at most

$$-\eta\log(m+1) + \sigma_m \log(2K_\star\varepsilon(B)) + \log(\sigma_m+1) + \log\frac{C_\star}{(r;r)_\infty^2}.$$

For fixed B the last three terms are $O_B(\sqrt{\log(m+1)} + \log\log(m+2))$. This proves the claim. No outer width is changed. □

The square-root logarithmic threshold is strictly smaller in order than the logarithmic-over-logarithmic threshold of the earlier factorial bound. The estimate depends on the actual two-end depth geometry, not on additional cancellation or a count of physical words.

14. Positive payment and the remaining low-load source

We now apply the joint-load theorem to the original localized persistent source. All quantities $\varepsilon(B), h_B, r_B, \gamma_B, K_B$ are fixed at the outer stage. A load level used below classifies an original trajectory at finite collision count; it is not a spectral, roof, residence or comparison window.

Fix $\eta > 0$ and set

$$u_m = (2+\eta)\log(m+1), \qquad \mathcal{Q}_{B,m} = \{\mathscr{L}_{\varepsilon(B)} \geq u_m\}. \tag{14.1}$$

Keep the previously paid set $\mathcal{Z}_{B,m}$ from (11.6). Define two restrictions of the same positive source, with no new renormalization:

$$\begin{aligned} \mathsf{U}_B^{\mathrm{load}} &= \mathsf{U}_B^{\mathrm{per}}[\mathcal{Z}_{B,m}^c \cap \mathcal{Q}_{B,m}], \\ \mathsf{U}_B^{\mathrm{low}} &= \mathsf{U}_B^{\mathrm{per}}[\mathcal{Z}_{B,m}^c \cap \mathcal{Q}_{B,m}^c], \\ \mathsf{R}_B^{\mathrm{low}} &= (\mathsf{U}_B^{\mathrm{low}} - K_B P_{r_B}^{\mathrm{ng}})_+. \end{aligned} \tag{14.2}$$

The comparison pool is the full original nongrazing pool, not a source-selected pool. The notation low refers only to incidence load; clearance witnesses remain at arbitrary depth.

Theorem 14.1 (A new collision-normalized positive payment)**.** *At every finite count,*

$$\mathsf{U}_B^{\mathrm{per}} = \mathsf{U}_B^{\mathrm{exc}} + \mathsf{U}_B^{\mathrm{load}} + \mathsf{U}_B^{\mathrm{low}}, \qquad 0 \leq \mathsf{R}_B^{\mathrm{loc}} - \mathsf{R}_B^{\mathrm{low}} \leq \mathsf{U}_B^{\mathrm{load}}. \tag{14.3}$$

For every fixed B, M, the load term has zero ordered height:

$$\mathcal{H}_M(\mathsf{U}_B^{\mathrm{load}}) = 0. \tag{14.4}$$

More explicitly, whenever $u_m > b_{\varepsilon(B)}$,

$$\sup_{R,n,k} m^2 \|\mathsf{U}_B^{\mathrm{load}}\|_\infty \leq C_\star \exp\left\{-\eta\log(m+1) + 2\sqrt{b_{\varepsilon(B)}(2+\eta)\log(m+1)}\right\}. \tag{14.5}$$

Consequently

$$\mathcal{H}_M(\mathsf{R}_B^{\mathrm{loc}} - \mathsf{R}_B^{\mathrm{low}}) = 0, \qquad \mathcal{E}_M \leq \mathcal{H}_M(\mathsf{R}_B^{\mathrm{low}}) + C_M B^{-1/384}. \tag{14.6}$$

The original assertion (1.5) *is equivalent to $\mathcal{H}_M(\mathsf{R}_B^{\mathrm{low}}) \to 0$ for every fixed M.*

Proof. The three sets in (14.3) disjointly partition the original persistent source. Thus its first equality holds before pushforward and remains exact afterwards. Monotonicity and the unit Lipschitz bound of the positive-part map prove the residual inequality with the same capacity on both sides.

Lemma 11.1 and Theorem 12.1 bound $\mathsf{U}_B^{\mathrm{load}}$ by $C_\star \exp[-u_m + 2\sqrt{b_{\varepsilon(B)}u_m}]$ once $u_m > b_{\varepsilon(B)}$. Multiplication by m^2, followed by $m^2 \leq (m+1)^2$, gives (14.5). At fixed B its exponent is $-\eta\log(m+1) + O_B(\sqrt{\log(m+1)})$, tending to minus infinity. The finitely many smaller counts do not affect the collision limsup. This proves (14.4) uniformly in all the stated inner suprema.

Apply the residual inequality and Corollary 11.3 to obtain (14.6). For necessity use $0 \leq \mathsf{R}_B^{\mathrm{low}} \leq \mathsf{U}_B^{\mathrm{micro}}$ and the original scalar equivalence. For sufficiency use (14.6) and then send B to infinity. This is the original order of limits. The optimizing parameter in the proof of (12.4) was chosen inside a finite-count inequality valid for every $0 < \lambda < 1$; it is not inserted into a theorem which holds only at a previously fixed outer width. □

Corollary 14.2 (Explicit support restrictions after payment)**.** *At any regular target point contributing to $\mathsf{U}_B^{\mathrm{low}}$, put $J = J_{\varepsilon(B)}$ and $N = |J|$. Then*

$$\alpha \sum_{i\in J} d_i < u_m, \qquad \alpha f(N) < u_m, \qquad \prod_{i\in J} \frac{c_i}{2\varepsilon(B)} > e^{-u_m}. \tag{14.7}$$

In particular $N < 2\sqrt{u_m/\alpha}$ and every individual witness has $c_i > 2\varepsilon(B)e^{-u_m}$. All these restrictions are in addition to the retained v111 incidence localization. They impose no clearance localization.

Proof. The load is strictly less than u_m on the retained set. Its nonnegative decomposition gives the first inequality. The sum of the depths of any N indices is at least $f(N)$, by the construction in Lemma 13.1; this proves the second. Exponentiating minus the definition of the load proves the product inequality. Since $f(N) \geq N^2/4$ and all load summands are nonnegative, the remaining assertions follow, including the empty-witness case. □

For sufficiently large B, the original section cap excludes incidence witnesses at the two endpoint collisions. The persistent guard then still forces an interior incidence witness or an original third-disk clearance witness. A state with $N = 0$ is not removed by this revision: the guard may be witnessed entirely by clearance. Critical entrances and section cuts carrying the original positive coefficient are likewise not deleted by their names.

The exact remaining target is

$$\lim_{B\to\infty} \limsup_{m\to\infty} \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} (\mathsf{U}^{\mathrm{low}}_{B,m}(t) - K_B P^{\mathrm{ng}}_{r_B,m}(t))_+ = 0. \tag{14.8}$$

Theorems 12.1–14.1 prove new finite-count concentration and positive payments in the required normalized topology. They do not prove (14.8). Low-load shallow incidences, arbitrary-depth clearance (including unvisited tangency), and capacity-deficient positive source remain in that equation. The original physical section, guard, arithmetic reference and exact-label pointwise assertion have not been weakened. The distinction between this remaining assertion and the proved payments is part of the theorem statement, not an inference from software diagnostics or an editorial change of topic.

15. Clearance-to-roof coarea without a fixed-clearance density bound

We keep the physical parameters and every event in Section 1. In this section u is the clearance coordinate of Section A, not physical time. Physical time is denoted by t throughout. Fix R, m, n, k, a flight $0 \leq j < m$, a sign σ, and an actual regular marked itinerary. On a local angular lift put

$$\mathcal{Q}_j(\vartheta, u) = (\alpha_0, \alpha_m)(X_{R,j}(\vartheta, \sigma, u)), \qquad S(\vartheta, u) = F(\mathcal{Q}_j(\vartheta, u)).$$

The point used as a mark is an unvisited near miss, not an additional reflection. Every derivative below is taken on the same designated regular itinerary. In particular m, j and the physical roof are not changed by this parametrization.

Theorem 15.1 (The long-record Jacobian and its critical Hessian)**.** *Let $A = D\mathcal{Q}_j$, let $g = \nabla_q F$, and let $G = D_q^2 F$. On every such chart,*

$$|\det A| = (R\omega)^{-1}, \tag{15.1}$$

$$|\det D_{(\vartheta,u)}(S, \arg g)| = \chi/(R\omega) \quad (g \neq 0), \tag{15.2}$$

$$D^2 S = A^{\mathsf{T}} G A + \sum_{a=1}^{2} g_a D^2 (\mathcal{Q}_j)_a. \tag{15.3}$$

At a regular critical roof, $g = 0$, the Hessian $D^2 S$ is positive definite and

$$\sqrt{\det D^2 S} = \frac{\sqrt{\det G}}{R\omega}. \tag{15.4}$$

These are identities for each finite physical collision count, with all coordinate and section-normalization factors displayed. In particular the Hessian term containing g cannot be dropped away from a critical roof.

Proof. The collision map preserves $d\alpha\, dp$ at fixed R. The signed line map at collision j has absolute Jacobian R, by (A.4). Composition with T_R^j therefore still has absolute Jacobian R with respect to the initial collision coordinates. The map $(\alpha_0, p_0) \mapsto (\alpha_0, \alpha_m)$ has absolute Jacobian ω^{-1} by the definition of ω. Dividing the two determinants gives (15.1). It uses the actual inverse iterate and is independent of where the mark occurs. On a noncritical chart the determinant of $q \mapsto (F, \arg g)$ is χ; the chain rule gives (15.2). Twice differentiating $F \circ \mathcal{Q}_j$ gives (15.3), component by component. At $g = 0$ its last term vanishes, and the retained positive-definite optical Hessian and the invertibility of A give the assertion. Taking the determinant and using (15.1) proves (15.4).

As a normalization check, the line-coordinate density is $(4\pi R c_*)^{-1}$. Dividing it by the determinant in (15.2) gives exactly $\omega/(4\pi c_* \chi)$, the original target-roof/Gauss-angle density. There is no additional factor R, R^{-1} or c_*^{-1}. All chart restrictions and exact labels remain on the source side of this change of variables. □

Corollary 15.2 (Uniform determinant growth on regular marked charts)**.** *With the retained constants $\varrho = 47/59$ and $\lambda_\circ$, all regular marked charts with the original endpoint cap satisfy*

$$|\det D\mathcal{Q}_j| \geq \frac{600}{2209}\varrho^{-(m-1)}, \qquad |\det D(S, \arg g)| \geq \frac{600\lambda_\circ}{2209}\varrho^{-(m-1)} \quad (g \neq 0).$$

At a critical roof the same lower bound as in the second inequality holds for $\sqrt{\det D^2 S}$. The constants are uniform in the original radius interval, the exact labels, the flight index, and the physical word. These determinant bounds do not bound individual singular values, a chart's condition number, or the complete physical sum of inverse determinants.

Proof. The retained optical estimate is $\omega \leq (47/6)\varrho^{m-1}$, and $R \leq 47/100$. Insert these in (15.1). Then use $\chi \geq \lambda_\circ$ and $\sqrt{\det G} \geq \lambda_\circ$ in (15.2) and (15.4). The identity $6/(47(47/100)) = 600/2209$ fixes the common constant. □

The following elementary estimate is recorded with proof to avoid importing an unstated finite-type or regularity theorem.

Lemma 15.3 (A one-dimensional band estimate)**.** *Let $f \in C^\ell(I)$ on an interval, with integer $\ell \geq 1$ and $|f^{(\ell)}| \geq \kappa > 0$. For $h > 0$,*

$$|\{x \in I : a < f(x) < a + h\}| \leq 2\ell(h/\kappa)^{1/\ell}. \tag{15.5}$$

The estimate also bounds any Borel subset of that band.

Proof. First restrict the band to a compact subinterval and let its measure be $L > 0$. Its cumulative Lebesgue measure is continuous and one-Lipschitz. The $\ell+1$ quantiles with successive mass gaps L/ℓ can be chosen in its closed support; their successive coordinate gaps are at least L/ℓ. Equivalently one can use interior approximate quantiles and pass to their limits. Write them as $x_0 < \cdots < x_\ell$. Continuity puts $f(x_i)$ in $[a, a + h]$. The divided-difference mean-value theorem, proved by subtracting the interpolating polynomial and applying Rolle's theorem ℓ times, gives some $\xi \in (x_0, x_\ell)$ for which

$$\frac{f^{(\ell)}(\xi)}{\ell!} = \sum_{i=0}^{\ell} \frac{f(x_i) - a}{\prod_{v \neq i}(x_i - x_v)}.$$

The subtraction of a is legitimate because the divided difference of a constant is zero. The gap bound makes the absolute denominator at least $(L/\ell)^\ell i!(\ell-i)!$. Hence

$$\kappa \le h(\ell/L)^\ell \sum_{i=0}^{\ell} \binom{\ell}{i} = h(2\ell/L)^\ell.$$

Rearrange, then exhaust the original interval by compact subintervals. This also handles a band not initially known to have finite measure. No statement about the number of physical words is used in the proof. □

Theorem 15.4 (A positive finite-type clearance slice)**.** *Suppose $S \in C^{\ell+1}$ on a neighbourhood of $I \times [0, \delta_0]$, and*

$$0 < b_f \le |S_u| \le B_f, \qquad |\partial_\vartheta^\ell S(\vartheta, 0)| \ge \kappa_f > 0.$$

The sign of S_u is fixed on this rectangle. Let E be any Borel subset of it and $0 \le h \le H_f$ a pointwise bounded Borel weight. For $0 < \delta \le \delta_0$, the measure

$$S_\#(h\mathbf{1}_E\mathbf{1}_{\{0<u<\delta\}}\, d\vartheta\, du)$$

has a density $D_{E,h,\delta}$ with

$$\|D_{E,h,\delta}\|_\infty \le \frac{2\ell H_f}{b_f}\left(\frac{B_f\delta}{\kappa_f}\right)^{1/\ell}. \tag{15.6}$$

In the physical marked record one takes $h = (4\pi R c_*)^{-1}$, or a bounded multiplier of this weight, and E is the original physical domain with all exact labels. The constants must be established on that actual chart; the theorem does not assert their uniformity in m or their summability over all charts.

Proof. At fixed ϑ, $u \mapsto S(\vartheta, u)$ is strictly monotone. Let $U_t(\vartheta)$ denote its unique inverse when it exists. The ordinary one-dimensional substitution, first against nonnegative roof tests and then by Tonelli, gives

$$D_{E,h,\delta}(t) = \int_I \frac{h(\vartheta, U_t)\mathbf{1}_E(\vartheta, U_t)\mathbf{1}_{\{0<U_t<\delta\}}}{|S_u(\vartheta, U_t)|}\, d\vartheta.$$

Missing inverses contribute zero. This argument proves absolute continuity, not merely an estimate conditional on its existence. If $S_u > 0$ and the inverse lies in $(0, \delta)$, then $t - B_f\delta < S(\vartheta, 0) < t$. For $S_u < 0$ the corresponding band is $t < S(\vartheta, 0) < t + B_f\delta$. Both have width $B_f\delta$. Bound the integrand by H_f/b_f and apply Lemma 15.3. Borel cuts, including section cuts, only restrict the integral and never create a chart-count multiplier. The same proof on bounded compact rectangles followed by monotone exhaustion gives the assertion for open intervals and Borel weights. □

Theorem 15.5 (Integrated simple folds, with a sharp local constant)**.** *Assume that on $I \times (0, \delta_0)$ one has $S_{\vartheta\vartheta} \ge a_f > 0$ and $S_u \ge b_f > 0$, and that for each u the unique angular critical point $\vartheta_0(u)$ exists in I. For the same positive weights and selectors as above,*

$$\|D_{E,h,\delta}\|_\infty \le 2H_f\sqrt{\frac{2\delta}{a_f b_f}}. \tag{15.7}$$

The assertion also applies after reversing the roof sign or the clearance-coordinate orientation when the corresponding derivative signs are reversed. It does not require a bounded roof density at each fixed clearance.

Proof. Put $v(u) = S(\vartheta_0(u), u)$. The implicit function theorem and the critical-point equation give $v'(u) = S_u(\vartheta_0(u), u) \geq b_f$. For fixed u there are at most two angular roots of $S(\vartheta, u) = t$, and

$$|S_\vartheta(\vartheta, u)|^2 \geq 2a_f(S(\vartheta, u) - v(u)).$$

Indeed differentiate S_ϑ^2 along each monotone side and integrate from its zero at ϑ_0. On the left side the integration orientation and the sign of S_ϑ reverse together, giving the same inequality. One-dimensional substitution on those two sides gives a fixed-u density bounded by $H_f\sqrt{2/a_f}(t - v(u))^{-1/2}$ for $t > v(u)$ and zero below. The angular critical point itself has zero angular measure. The bound is allowed to be infinite at $t = v(u)$.

For $0 < u < \delta$, if t is in the range of v, let u_t be its unique inverse. The contributing values satisfy $u < u_t$, and $t - v(u) \geq b_f(u_t - u)$. If $t \geq v(\delta)$, replace u_t by δ for an upper bound; if $t \leq v(0)$ there is no contribution. In every case

$$\int_0^\delta \mathbf{1}_{\{t>v(u)\}}(t - v(u))^{-1/2}\, du \leq 2\sqrt{\delta/b_f}.$$

Integrate the fixed-u estimate by Tonelli to obtain (15.7). Restricting by E preserves the inequality. For $S(\vartheta, u) = \vartheta^2 + u$, $h = 1$, and the whole real angular line, the integrated density is

$$2\{\sqrt{t_+} - \sqrt{(t-\delta)_+}\},$$

whose maximum is $2\sqrt{\delta}$. Thus the constant is attained when $a_f = 2$ and $b_f = 1$. This elementary normal form checks the local analytic bound only; it is not substituted for a billiard trajectory or for the joint-record distribution. □

15.1. **One-copy summation and the remaining uniformity.** On a countable collection of admissible physical charts, first replace their source sets by disjoint Borel differences. The preceding density formulae then add exactly. Their positive upper bounds add as well, with an extended value permitted until a finite bound has been proved. Two impact signs and distinct unvisited centres are retained with precisely the multiplicity in (D.2); no full-word cardinality is inserted. Borel event cuts do not require a new chart merely because they disconnect the selected set.

In particular the per-clearance condition (D.3) is not needed on a chart controlled by Theorem 15.5. An angular fold may have an unbounded conditional density and a bounded integrated density at the same time. What is still required for the full clearance payment is a bound for the complete, exact-label, positively weighted chart sum, in the original $m^2L^\infty(dt)$ topology. The local constants in (15.6) or (15.7) have not been proved to provide that bound. Charts with $S_u = 0$, degenerate angular behaviour outside a verified finite type, and singular physical boundaries are not discarded. Regular full critical roofs are treated next by a different exact calculation.

16. Critical roofs, actual reversing orbits, and their exact density coefficient

We now identify a dynamical invariant in the critical-roof density, without averaging the roof or replacing the physical trajectories. Let x be a regular m-flight trajectory with endpoints in the original section cap. Write $q_x = (\alpha_0, \alpha_m)$ and $t_x = F(q_x)$. All statements about derivatives of the actual collision map below concern regular trajectories, not a formal continuation through a tangent collision.

Theorem 16.1 (Critical half-paths are physical reversing periodic orbits)**.** *The endpoint roof is critical precisely when $p_0 = p_m = 0$. Such a trajectory extends by the actual reflection law to a regular $2m$-collision periodic orbit, whose period need not be primitive. The extra m flights*

traverse the original flights in reverse, with total roof $2t_x$ and total lifted displacement zero. Let $\mathcal{I}(\alpha,p)=(\alpha,-p)$ be collision time reversal, and put

$$A^{\mathcal{I}}_{m,R}(x)=\sum_{j=0}^{m-1}\mathbf{1}_{Y^*_R}(\mathcal{I}T_R^j x).$$

If the half-path belongs to the original exact event, then

$$A_{2m,R}(x)=A_{m,R}(x)+A^{\mathcal{I}}_{m,R}(x). \tag{16.1}$$

One cannot replace the right side by $2n$ without additionally verifying the time-reversed section tests.

Proof. The first-variation identity already retained in (2.1) is $\nabla_q F=R(-p_0,p_m)$, which proves the criticality assertion. At a normal terminal collision the reflection sends the incoming velocity to its negative. The ray therefore retraces the preceding actual flight. At each intervening regular collision the reversed reflection retraces the next preceding flight. At the normal initial collision it returns to the original outgoing state. Thus every reversed flight is an actual free flight, with no extra or hidden collision, and the orbit closes after $2m$ collisions. Its reversed displacement is the negative of the forward displacement. Its length is twice the original length. No assertion about a primitive period is used.

For $1\le j<m$ the reversed state at time $2m-j$ is $\mathcal{I}T_R^j x$, while the states at times 0 and m are fixed by $\mathcal{I}$. In the original event both endpoint section tests equal one. Summing the m forward tests and then the terminal test and the $m-1$ reversed interior tests gives (16.1), since the initial test in $A^{\mathcal{I}}_m$ also equals one. The original section is not assumed invariant under $\mathcal{I}$. □

Theorem 16.2 (Exact stability determinant at a critical roof)**.** *At a regular critical half-path write*

$$G_x=\begin{pmatrix}a & e\\ e & d\end{pmatrix},\qquad \Delta_x=ad-e^2>0,\qquad \mathcal{M}_x=DT_R^{2m}(x).$$

Then $e\neq 0$, and the actual return derivative satisfies

$$\det\mathcal{M}_x=1,\qquad \operatorname{tr}\mathcal{M}_x-2=|\det(I_2-\mathcal{M}_x)|=\frac{4\Delta_x}{e^2}>0. \tag{16.2}$$

Consequently the critical density coefficient

$$\mathfrak{J}(x):=\frac{\omega(x)}{2c_*\sqrt{\det G_x}}=\frac{1}{Rc_*\sqrt{|\det(I_2-DT_R^{2m}(x))|}} \tag{16.3}$$

is an invariant of the actual reversing periodic orbit with its marked initial state and specified half-length m. The coefficient is not a probability conditioned on that periodic orbit.

Proof. The signed optical corner formula (F.1) makes e nonzero on every regular finite trajectory. Differentiate $p_0=-F_{\alpha_0}/R$ and $p_m=F_{\alpha_m}/R$. Solving the first identity for $d\alpha_m$ gives, in the original (α,p) coordinates,

$$B_x:=DT_R^m(x)=\begin{pmatrix}-a/e & -R/e\\ -\Delta_x/(Re) & -d/e\end{pmatrix},\qquad \det B_x=1. \tag{16.4}$$

This keeps the sign of e; replacing it prematurely by $|e|$ in the matrix would change a derivative, although not its final trace. At both normal endpoints the derivative of time reversal is $J=\operatorname{diag}(1,-1)$. The reversibility identity $\mathcal{I}T_R^m\mathcal{I}=T_R^{-m}$ therefore yields $DT_R^m(T_R^m x)=JB_x^{-1}J$ and

$$\mathcal{M}_x=JB_x^{-1}JB_x.$$

For an arbitrary determinant-one matrix $B = \left(\begin{smallmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{smallmatrix}\right)$, $JB^{-1}J = \left(\begin{smallmatrix} b_{22} & b_{12} \\ b_{21} & b_{11} \end{smallmatrix}\right)$, so

$$\operatorname{tr}(JB^{-1}JB) - 2 = 4b_{12}b_{21}.$$

Substitution of (16.4) gives $4\Delta_x/e^2$. For a two-dimensional determinant-one matrix, $\det(I_2 - \mathcal{M}_x) = 2 - \operatorname{tr}\mathcal{M}_x$. This proves (16.2) and also gives positive real hyperbolic multipliers. Finally $\omega = |e|/R$, so elementary simplification proves (16.3). The determinant of $I_2 - \mathcal{M}_x$ is invariant under a smooth change of collision coordinates. No periodic-orbit trace formula or distributional limit theorem has been invoked. □

Theorem 16.3 (Critical-roof coarea with arbitrary exact-event cuts)**.** *Fix a regular critical half-path and a sufficiently small endpoint neighbourhood V on its designated itinerary. Let $E \subset V$ be an arbitrary Borel selector, including the actual physical domain, initial and terminal section tests, exact occupation and displacement labels, and any clearance tests. Put*

$$y = G_x^{1/2}(q - q_x), \quad f(y) = F(q) - t_x, \quad w(y) = \frac{\omega(q)}{4\pi c_* \sqrt{\det G_x}}.$$

Choose $r_0 > 0$ so that the closed y-ball is contained in the coordinate neighbourhood, and constants $K, L_w \geq 0$ such that

$$\|D^2 f(y) - I_2\| \leq K|y|, \quad |w(y) - w(0)| \leq L_w|y|, \quad Kr_0 \leq 1.$$

For $0 < s < r_0^2/3$ there is, for every angle θ, a unique $r_s(\theta) \in (0, r_0)$ with $f(r_s e(\theta)) = s$. Define

$$a_E(s) = \frac{1}{2\pi} \int_0^{2\pi} \mathbf{1}_E\big(q_x + G_x^{-1/2} r_s(\theta) e(\theta)\big)\, d\theta.$$

The original positive collision measure restricted to E and this ball has a density D_E near t_x, no atom at t_x, and

$$|D_E(t_x + s) - \mathfrak{J}(x) a_E(s)| \leq 2\pi\sqrt{3}\{2L_w + Kw(0)\}\sqrt{s}. \tag{16.5}$$

There is no contribution below t_x. The constant is independent of the Borel selector E, but is a constant of this local chart, not a claimed uniform-in-m constant.

If E contains a neighbourhood of q_x, then $a_E(s) = 1$ for small s and the one-sided density limit is exactly $\mathfrak{J}(x)$. If the selected physical side is described by finitely many C^1 inequalities vanishing at q_x with nonzero linear parts, then its limiting coefficient is $\mathfrak{J}(x)$ times the angular fraction of the corresponding linear sector. Without such a hypothesis the exact $a_E(s)$ is retained; no limit for it is asserted.

Proof. The retained reduced action is smooth on a regular finite itinerary; the inverse function theorem and the nonzero optical corner give a smooth positive ω there. Thus the displayed constants exist after decreasing r_0. At a critical point $f(0) = 0$, $\nabla f(0) = 0$ and $D^2 f(0) = I_2$. Integrating the Hessian bound along a radial segment gives

$$|\partial_r f(re) - r| \leq Kr^2/2, \qquad |f(re) - r^2/2| \leq Kr^3/6.$$

For $r \leq r_0$ these imply $r^2/3 \leq f(re) \leq 2r^2/3$, $\partial_r f \geq r/2$, and hence the existence and uniqueness of r_s, with $r_s \leq \sqrt{3s}$. They also give

$$\left|\frac{r_s}{\partial_r f(r_s e)} - 1\right| \leq Kr_s, \qquad 0 < \frac{r_s}{\partial_r f(r_s e)} \leq 2.$$

In endpoint coordinates the original collision density is $\omega(q)/(4\pi c_*)$; the linear substitution to y makes it $w(y)\, dy$. Polar coordinates and the monotone radial substitution $t = t_x + f(re(\theta))$, tested against nonnegative roof functions, give the exact density

$$D_E(t_x + s) = \int_0^{2\pi} \mathbf{1}_E(q_s(\theta)) w(r_s e(\theta)) \frac{r_s}{\partial_r f(r_s e(\theta))}\, d\theta.$$

Here $q_s(\theta) = q_x + G_x^{-1/2} r_s e(\theta)$, and any source outside the selected ball is not part of D_E. The point $r = 0$ has zero two-dimensional measure, so it produces no atom. The strict radial positivity also excludes mass below t_x inside the ball. For each angle the difference between the weight in this integral and $w(0)$ is at most $(2L_w + Kw(0))r_s$. Integrating gives (16.5), since $2\pi w(0) = \mathfrak{J}(x)$. This estimate does not differentiate an indicator or require its boundary to have zero roofwise length.

When E contains a neighbourhood, every sufficiently small radial point belongs to it. For the stated finite collection of C^1 cuts, Taylor expansion of each defining function shows convergence of its sign to its nonzero linear part at every angle except the finitely many zero directions. Dominated convergence gives the sector fraction. It is the fraction after the transformation by $G_x^{-1/2}$, not an unweighted fraction in the original endpoint angles. For a general Borel selector no such argument is used. □

Corollary 16.4 (One-sided unvisited boundaries without a fictitious orbit)**.** *Suppose a designated finite broken path reaches an unvisited-disk tangency or a section cut, all its* visited *incidences remain positive, and its endpoint action is critical. The sector formula* (16.5)*, with the original physical-side selector, continues to hold in the one-sided coordinate neighbourhood. At a singular state the second expression in* (16.3) *is* not *asserted to be a derivative of a defined physical return map; the first expression is the one-sided action coefficient.*

Proof. The stationary equations for the visited interior collision points have positive-definite interior Hessian. At the stated limiting path they therefore have a smooth local solution as functions of the two endpoints. The signed optical corner is nonzero because the visited incidences and flight lengths are positive. This supplies the smooth action and endpoint-coordinate Jacobian on a full auxiliary neighbourhood. The portion representing actual trajectories is retained as a Borel subset E; portions intersecting a forbidden disk are never assigned to the physical event. On the admitted side these functions agree with the actual action and original measure. The proof of Theorem 16.3 used arbitrary Borel restrictions, so it applies to this subset without continuing the physical dynamics across its boundary. Simultaneous cuts are included in the same selector. If visited grazing also occurs, this particular smooth extension is not invoked. Its treatment remains the inherited visited-grazing trace argument. Nothing in this corollary equates an area-null boundary point with a zero positive pairing-endpoint trace. □

Proposition 16.5 (A complete finite-count critical coefficient bound)**.** *Let the sum run over all distinct regular critical half-paths in the original exact event, at fixed m, R, n, k. Then*

$$\mathfrak{J}(x) \le \frac{47}{12 c_* \lambda_\circ} \varrho^{m-1}. \tag{16.6}$$

Under the retained, explicitly stated geometric-pressure input 10.1, the full positive sum satisfies

$$\sum_x \mathfrak{J}(x) \le \frac{C_0 Q_*}{2 c_* \lambda_\circ}. \tag{16.7}$$

This is unscaled. It does not supply an m^{-2} bound for a clearance-selected critical source or a uniform bound for the local error constants in (16.5)*.*

Proof. Use $\sqrt{\det G_x} \ge \lambda_\circ$ in (16.3) and the optical bound for ω in (3.8) to obtain (16.6). The retained global equal-gradient uniqueness gives at most one regular critical half-path per lifted physical word. Such a path extends periodically without singular collisions, so it belongs to the two-sided regular set M_R'. The selected-word sum in (10.3), with no incidence marks, therefore bounds $\sum_x \omega(x)$ by $C_0 Q_*$. Divide by $2c_*\lambda_\circ$. Missing words contribute zero. No multiplication of a per-orbit exponential estimate by the number of words is made. □

16.1. **What the critical calculation does and does not close.** Formula (16.3) identifies an actual return-stability weight in the original roof density; it is not an ansatz for a Markov model and not a periodic-orbit replacement of the invariant measure. Formula (16.5) also keeps positive source on section-cut and one-sided unvisited boundaries rather than assigning it a zero coefficient by area-nullness. The full paired source, its nongrazing comparison pool, and (14.8) are unchanged. A critical *component* coefficient need not equal a discontinuity of the entire density, since other physical components may be present at the same roof.

For a collision-uniform payment it is still necessary to control the actual weighted sum on fixed outer neighbourhoods, its clearance and exact-label localization, and the chart remainders. Choosing a separate collision-dependent neighbourhood merely to separate critical roof values would not prove that fixed-window statement. Neither (16.6) nor the unscaled sum (16.7) is used in that invalid manner. The new identities instead give the exact geometric and stability coefficients that a positive capacity or transfer-operator estimate must control.

17. Exact two-mark clearance flux on one physical flight

The preceding clearance–roof analysis treats a selected near miss as a *mark*, without adding a collision. Two different unvisited disks can be marked on the same actual flight. Here we quantify their physical multiplicity exactly, rather than assuming independence or multiplying by the number of collision words. The result is a second-order *collision-space* clearance theorem; it is not an m^2-normalized pointwise roof estimate.

Throughout let $R \in [R_-, R_+]$, $R_+ = 47/100$, and keep $\delta_* = 3/200$, $0 < \delta_1, \delta_2 \le \delta_*$, the triangular lattice Λ, the actual flight time $\tau \le \tau_+$, and the one-copy physical mark maps X_R, Z_R of Theorem A.2. Define the strictly positive *physical separation* and the finite comparison set

$$\begin{aligned} c_{\mathrm{pair}} &= \sqrt{1 - [2(R_+ + \delta_*)]^2} = \frac{\sqrt{591}}{100} > 0, \\ \mathcal{W}_* &= \{w \in \Lambda \setminus \{0\} : |w| \le \tau_+ + 2(R_+ + \delta_*)\}, \\ N_{\mathrm{pair}} &= |\mathcal{W}_*|. \end{aligned} \tag{17.1}$$

The constant N_{pair} is finite by the *same* finite-horizon assumption already used in Lemma A.1; it is not the number of admissible words, the number of pairing sheets, or an unproved distortion constant.

Lemma 17.1 (Distinct same-flight clearances are uniformly transverse)**.** *Let $z_1 \ne z_2$ be two unvisited disk centers whose clearance values $u_i \in (0, \delta_i)$ are realized by the* same *regular physical flight with unit direction v_ϑ. Let $b_i = (z_i - q) \cdot v_\vartheta^\perp$ be their signed impact parameters. Then both perpendicular feet lie in the open physical flight, $z_2 - z_1 \in \mathcal{W}_*$, and*

$$|(z_2 - z_1) \cdot v_\vartheta| = \sqrt{|z_2 - z_1|^2 - (b_2 - b_1)^2} \ge c_{\mathrm{pair}}. \tag{17.2}$$

In particular, for the first signed impact parameter held fixed, $\partial_\vartheta b_2 = -(z_2 - z_1) \cdot v_\vartheta$ does not vanish on a simultaneous physical two-mark state.

Proof. Lemma A.1 puts each foot in the open physical flight. Thus its along-flight coordinate lies in an interval of length at most τ_+. Since $|b_i| = R + u_i$,

$$|b_2 - b_1| \le 2R + \delta_1 + \delta_2 \le 2(R_+ + \delta_*) = 97/100 < 1.$$

The lattice spacing $|z_2 - z_1| \ge 1$ and orthogonal decomposition along $v_\vartheta, v_\vartheta^\perp$ imply the equality and lower bound in (17.2). The same decomposition gives $|z_2 - z_1| \le \tau_+ + 2(R_+ + \delta_*)$, proving membership in $\mathcal{W}_*$. The impact derivative follows from $dv_\vartheta^\perp/d\vartheta = -v_\vartheta$ with b_1 fixed. Neither a grazing endpoint nor an unvisited near miss is treated as an extra collision. □

For a first center translated to 0, fix $w \in \mathcal{W}_*$ and $\sigma_1, \sigma_2 \in \{\pm 1\}$. Write

$$d_{\sigma_1\sigma_2}(u_1, u_2) = \sigma_2(R + u_2) - \sigma_1(R + u_1). \tag{17.3}$$

For all $0 < u_i < \delta_i$, $|d_{\sigma_1\sigma_2}| < 97/100 < |w|$. Thus the equation

$$w \cdot v_\vartheta^\perp = d_{\sigma_1\sigma_2}(u_1, u_2) \tag{17.4}$$

has exactly two roots modulo 2π, denoted $\vartheta_{w,\sigma_1,\sigma_2,+}(u_1, u_2)$ and $\vartheta_{w,\sigma_1,\sigma_2,-}(u_1, u_2)$, distinguished by the sign of $w \cdot v_\vartheta$. They are smooth on each parameter rectangle. Their one-dimensional change-of-variables Jacobian is

$$\left|\frac{\partial\vartheta}{\partial u_2}\right| = \frac{1}{\sqrt{|w|^2 - d_{\sigma_1\sigma_2}(u_1, u_2)^2}}, \tag{17.5}$$

with u_1 fixed; the sign of σ_2 has unit magnitude.

Theorem 17.2 (An exact physical two-mark coarea identity)**.** *For every nonnegative Borel weight $a(x, z_1, z_2)$ of a genuine outgoing collision state and two* ordered*, distinct unvisited centers (expressed in the initial incident disk's lattice frame), set at a root of* (17.4)

$$x = X_R(\vartheta, \sigma_1, u_1), \quad z_1 = Z_R(\vartheta, \sigma_1, u_1), \quad z_2 = z_1 + w, \quad \mathfrak{A} = \mathbf{1}_{\{z_2 \in \mathcal{Z}_{\delta_2}(x)\}}.$$

Then the full, one-copy *original collision measure obeys*

$$\begin{aligned} &\int \sum_{z_1 \in \mathcal{Z}_{\delta_1}(x)} \sum_{\substack{z_2 \in \mathcal{Z}_{\delta_2}(x) \\ z_2 \neq z_1}} a(x, z_1, z_2)\, d\nu(x) \\ &\quad = \frac{1}{4\pi R} \sum_{w \in \mathcal{W}_*} \sum_{\sigma_1,\sigma_2,\eta \in \{\pm 1\}} \int_0^{\delta_1} \int_0^{\delta_2} \left. \frac{\mathfrak{A}\, a(x, z_1, z_1 + w)}{\sqrt{|w|^2 - d_{\sigma_1\sigma_2}(u_1, u_2)^2}} \right|_{\vartheta = \vartheta_{w,\sigma_1,\sigma_2,\eta}} du_2\, du_1. \end{aligned} \tag{17.6}$$

The mask $\mathfrak{A}$ retains both actual feet*, the same physical collision flight, and the prohibition on unvisited-disk reflections. It must not be removed in an equality. Any original return, displacement, collision, roof, section, or radius test may be included in a as a Borel selector;* (17.6) *never modifies that test or the physical clock.*

Proof. Apply the proved one-mark flux identity (A.2) to the first mark z_1. This gives the factor $(4\pi R)^{-1}$, the sign σ_1, the u_1 integral and the original angular integral. For each (ϑ, u_1), translating z_1 to the origin associates any distinct second center with exactly one $w = z_2 - z_1 \in \mathcal{W}_*$ by Lemma 17.1. The second center has a definite sign σ_2 of its impact and a unique $u_2 \in (0, \delta_2)$; its physical membership is precisely $\mathfrak{A} = 1$. For fixed w, σ_1, u_1, the function $\vartheta \mapsto b_2 = b_1 + w \cdot v_\vartheta^\perp$ is a sine wave around the fixed b_1. On each of the two signed impact bands $\sigma_2(R, R + \delta_2)$, equation (17.4) has exactly two roots, and its derivative has the nonzero magnitude in (17.5). One-dimensional substitution on the two monotone branches therefore gives the sum over η and the displayed denominator. The total sum over w is finite; Tonelli and monotone approximation first for bounded weights then for arbitrary nonnegative Borel a justify all substitutions. Only regular charts of the genuine flight are used; the inherited one-mark construction assigns a fixed null-set convention on singular lines. The formula counts an ordered pair once because its first mark is unique in the first flux identity and its relative w and sign/angle root are unique thereafter. No physical word count, symbolic renewal process, conditional independence, or fictitious collision is introduced. □

Theorem 17.3 (Uniform quadratic bound for two distinct unvisited marks)**.** *For the original invariant collision measure and every pair of allowed clearances,*

$$\int \sum_{z_1\in\mathcal{Z}_{\delta_1}(x)} \sum_{\substack{z_2\in\mathcal{Z}_{\delta_2}(x)\\ z_2\neq z_1}} 1\, d\nu(x) \le \frac{2N_{\mathrm{pair}}}{\pi R c_{\mathrm{pair}}}\,\delta_1\delta_2 \le \frac{2N_{\mathrm{pair}}}{\pi R_- c_{\mathrm{pair}}}\,\delta_1\delta_2. \tag{17.7}$$

This bound includes every physical line/flight and both signs; it does not require independence between the two marks.

Proof. Apply Theorem 17.2 with $a \equiv 1$. Drop only the positive *mask in the upper bound.* The denominator is at least c_{pair} by the lattice-spacing calculation, and there are exactly N_{pair} candidate differences and $2^3 = 8$ sign/root choices. Thus the right side is at most $8N_{\mathrm{pair}}\delta_1\delta_2/(4\pi R c_{\mathrm{pair}})$, which is (17.7). The denominator lower bound is pointwise on the entire parameter rectangle, not only almost everywhere on the physically admitted pairs. □

Corollary 17.4 (A sharp first-order unique-clearance law)**.** *Let* $N_\delta = |\mathcal{Z}_\delta|$ *be the* original *one-flight unvisited-near-miss count and write* $C_{\mathrm{pair}} = N_{\mathrm{pair}}/(\pi R_- c_{\mathrm{pair}})$*. Then*

$$0 \le \frac{\delta}{R} - \nu(N_\delta \ge 1) \le C_{\mathrm{pair}}\delta^2, \tag{17.8}$$

$$0 \le \nu(N_\delta \ge 2) \le C_{\mathrm{pair}}\delta^2, \tag{17.9}$$

$$\nu(N_\delta = 1) \ge \frac{\delta}{R} - 2C_{\mathrm{pair}}\delta^2. \tag{17.10}$$

In particular $\nu(N_\delta \ge 1) = \delta/R + O(\delta^2)$*, uniformly in the original radius interval; the* δ/R *term is exactly the one-copy physical flux, not a proposed Poisson law.*

Proof. For every nonnegative integer N, $0 \le N - \mathbf{1}_{\{N\ge1\}} = (N-1)_+ \le N(N-1)/2$ and $\mathbf{1}_{\{N\ge2\}} \le N(N-1)/2$. Also $N\mathbf{1}_{\{N\ge2\}} \le N(N-1)$. Apply these deterministic inequalities to N_δ, use the exact physical mean (A.3), and substitute Theorem 17.3 with $\delta_1 = \delta_2 = \delta$. No limit in collision count is needed for these one-flight statements. □

Corollary 17.5 (Original exact-label records: uniform weak overcount payment)**.** *Fix* m, R, n, k*, the original physical* m*-flight record, and a nonnegative Borel roof test* $0 \le \psi \le 1$*. Let* $0 < \delta_j \le \delta_*$ *for* $0 \le j < m$*. With the original* $\Pi = \Pi_{n,k,m,R}$ *and unchanged normalized source* ν_R^**,*

$$0 \le \int \mathbf{1}_\Pi \psi(L) \sum_{j=0}^{m-1} \left[N_{\delta_j}(T_R^j x) - \mathbf{1}_{\{N_{\delta_j}(T_R^j x)\ge1\}}\right] d\nu_R^*(x) \tag{17.11}$$

$$\le \frac{C_{\mathrm{pair}}}{c_*} \sum_{j=0}^{m-1} \delta_j^2.$$

For the original depth schedule $\delta_j = 2\varepsilon r^{d'_j}$*,* $0 < 2\varepsilon \le \delta_*$*,* $r = q_0^{1/4}$*, this is at most*

$$\frac{8C_{\mathrm{pair}}\varepsilon^2}{c_*(1-r^2)}, \tag{17.12}$$

uniformly in m, R, n, k *and the test* ψ*. Thus the positive roof measure of within-flight* duplicate marking *has uniform total variation of order* ε^2*, even with the full original exact labels. The corresponding density need not have controlled* L^∞ *height: multiplying this mass estimate by* m^2 *at fixed outer* B *is* not *justified.*

Proof. The integrand in (17.11) is nonnegative. Since $\Pi \subset Y_R^*$ and $\nu_R^* = c_*^{-1}\mathbf{1}_{Y_R^*}\nu$, its integral is bounded by c_*^{-1} times the sum of unconditioned collision integrals with $\psi \le 1$. By invariance

of the original collision measure, each integral equals the same one-flight quantity at $T_R^j x$ and is bounded by $C_{\mathrm{pair}}\delta_j^2$ from Corollary 17.4. The depth list has at most two indices at each nonnegative depth, so $\sum_j \delta_j^2 \le 2(2\varepsilon)^2 \sum_{d\ge 0} r^{2d} = 8\varepsilon^2/(1-r^2)$. All source and time tests were retained until applying positive domination; no independent probability factor and no exchange of the original limits occurs. □

The exact second-order formula (17.6) isolates the nontransverse configurations which a two-mark pointwise analysis would otherwise need to pay. The separation constant $c_{\mathrm{pair}} > 0$ rules out that particular simultaneous same-flight angle degeneracy on the original radius range. This leaves *single* unvisited-disk clearance, marks on different physical flights, grazing/section/pairing boundary terms, and the nongrazing capacity shortage. In particular the full $\mathcal{H}_M(\mathsf{R}_B^{\mathrm{low}})$ limit and the original raw pointwise local limit theorem remain open; the new $O(\varepsilon^2)$ total variation gain is never substituted for the required $m^2 L^\infty$ estimate.

Appendix A. An exact flux identity for unvisited near misses

The incidence load does not see a trajectory whose original guard is witnessed only by clearance. We introduce a counting observable on the *same* physical flights, not a new collision map. Throughout this section let

$$g_* = 1 - 2R_+ = 3/50, \qquad \delta_* = g_*/4 = 3/200, \qquad 0 < \delta \le \delta_*.$$

The symbol δ_* here is a one-flight geometric constant, not the outer microscopic width $\delta(B)$. All disks still have radius R.

For a regular outgoing collision x, write its flight as $q(x) + tv(x)$, $0 \le t \le \tau(x)$. Let $\mathcal{Z}_\delta(x)$ be the set of lattice centers other than the two incident centers for which

$$0 < \min_{0\le t\le \tau(x)} |q(x) + tv(x) - z| - R < \delta, \qquad N_\delta(x) = |\mathcal{Z}_\delta(x)|.$$

This counts near misses, not collisions. In particular neither the collision index, the physical roof, nor the section-visit count is incremented by N_δ. A center is counted once on a flight. Tangencies may be assigned value zero on their collision-area null set; no assertion about a positive pairing-endpoint trace follows from this convention.

Lemma A.1 (An interior foot and uniform endpoint separation)**.** *Every center counted by N_δ has its perpendicular foot in the open physical flight. If t_z is the time of that foot, then*

$$t_z \ge g_* - \delta, \qquad \tau - t_z \ge g_* - \delta. \tag{A.1}$$

The candidate centers, after translating the initial incident center to zero, belong to the finite set

$$\{z \in \Lambda \setminus \{0\} : |z| \le \tau_+ + 2R_+ + \delta_*\}.$$

Write N_* for its cardinality. This constant is independent of the itinerary length and of all exact record labels.

Proof. At either incident point, the distance to a third center is at least $1-R$, since distinct lattice centers have distance at least one. Its distance to the third disk is therefore at least $1 - 2R \ge g_*$. A minimum below $\delta \le g_*/4$ cannot occur at an endpoint. The nearest point on a line segment is then its perpendicular foot, whose distance from z is at most $R + \delta$. By the reverse triangle inequality its distance along the line from either endpoint is at least $(1 - R) - (R + \delta) \ge g_* - \delta$. Finally a foot is at distance at most $R + \tau_+$ from the initial center. Adding its distance from z gives the finite-set bound. This proof uses the actual segment, including both of its endpoints, rather than an infinite-line substitute for the clearance event. □

Put $v_\vartheta = (\cos\vartheta, \sin\vartheta)$ and $v_\vartheta^\perp = (-\sin\vartheta, \cos\vartheta)$. For a mark z define the signed line coordinate $b = (z-q)\cdot v_\vartheta^\perp$. Translate the marked center to zero. For $\sigma \in \{-1,1\}$ and $0 < u < \delta_*$ the oriented line with $b = \sigma(R+u)$ has closest point $-\sigma(R+u)v_\vartheta^\perp$. Follow this line backwards to its last actual collision. Reduce that collision by a lattice translation to the initial fundamental disk, and denote the resulting outgoing state and marked-center displacement by

$$X_R(\vartheta,\sigma,u), \qquad Z_R(\vartheta,\sigma,u).$$

These maps use straight free flight until an *actual* disk is met. There is no reflection at the closest point. The finite-horizon assumption ensures actual collisions in both directions, apart from null singular parameters on which these maps can be defined arbitrarily. The interval from the preceding collision to the following one remains the original physical flight.

Theorem A.2 (One-copy clearance flux and exact multiplicity)**.** *For every nonnegative Borel function a of an outgoing state and a marked-center displacement,*

$$\int \sum_{z\in\mathcal{Z}_\delta(x)} a(x,z)\,d\nu(x) = \frac{1}{4\pi R}\sum_{\sigma=\pm1}\int_0^{2\pi}\int_0^{\delta} a(X_R(\vartheta,\sigma,u), Z_R(\vartheta,\sigma,u))\,du\,d\vartheta. \tag{A.2}$$

In particular,

$$\int N_\delta\,d\nu = \delta/R. \tag{A.3}$$

Both statements hold for every $R \in [R_-, R_+]$. The multiplicity on the left is the number of physically distinct unvisited disks, not an upper bound for the number of collision words.

Proof. We first verify the local measure, including its radius factor. Write $q = R(\cos\alpha, \sin\alpha)$ and $\vartheta = \alpha + \arcsin p$. At fixed ϑ,

$$\left.\frac{\partial b}{\partial\alpha}\right|_\vartheta = -R\cos(\vartheta-\alpha), \qquad \frac{\partial\vartheta}{\partial p} = (1-p^2)^{-1/2}.$$

Consequently

$$\left|\det D_{(\alpha,p)}(\vartheta,b)\right| = R, \qquad d\nu = \frac{d\vartheta\,db}{4\pi R}. \tag{A.4}$$

The two signs $b = \sigma(R+u)$ both have $|db| = du$.

There is also a global multiplicity assertion to prove. A marked physical flight maps to its unique oriented line relative to its marked center. Conversely, the closest point of every line just described is in the free region: its distance to any other lattice center is at least $1 - R - u > R$. In fact the disk of radius $R+\delta_*$ around the marked center does not intersect any other actual disk, since $2R+\delta_* < 1$. The central disk is missed because the line has impact magnitude greater than R. Following the line in both directions therefore gives a physical flight whose incident disks are different from the marked disk. Its foot is interior and its clearance from that disk is u. This is inverse to the marked-flight map, modulo the prescribed lattice translation. Multiple near misses on one flight give different marked copies; they are neither lost nor identified.

For measure-theoretic precision, partition the line parameters by the actual backward incident disk and the forward incident disk, and exclude their tangent boundaries. These partitions are Borel; on each regular piece the disk-intersection formulas and their inverses are smooth. Tangency boundaries satisfy the line equation $|b_z| = R$. For any fixed alternative center their equation in (ϑ, u) is a graph in u (with one of the two fixed signs), hence has planar Lebesgue measure zero. There are countably many centers. The corresponding singular collision sets are also null by (A.4) on regular charts. Exhaust these charts by relatively compact subcharts and disjointize them. Applying the local Jacobian formula to bounded functions on finite unions, then using monotone convergence for both chart unions and $a \wedge n$, proves (A.2) without a prior integrability assumption. Setting $a = 1$ gives $2(2\pi)\delta/(4\pi R) = \delta/R$. □

Corollary A.3 (All physical depths with the original normalization)**.** *For arbitrary* $0 < \delta_j \le \delta_*$, $0 \le j < m$,

$$\int \sum_{j=0}^{m-1} N_{\delta_j}(T_R^j x)\, d\nu(x) = \frac{1}{R} \sum_{j=0}^{m-1} \delta_j.$$

For every original exact event Π *the same integral with* $\mathbf{1}_\Pi\, d\nu_R^*$ *is at most* $(Rc_*)^{-1} \sum_j \delta_j$*. In particular, if* $2\varepsilon \le \delta_*$ *and* $r = q_0^{1/4}$*, then*

$$\nu_R^*\Big(\Pi \cap \{\exists j : \Delta_j < 2\varepsilon r^{d'_j}\}\Big) \le \frac{4\varepsilon}{R_- c_*(1-r)}. \tag{A.5}$$

Proof. The collision measure ν is invariant under the original elastic collision map. This also follows from the oriented-line Jacobian (A.4), whose magnitude is the same at both incident circles, and specular reflection, which preserves boundary flux. Thus apply (A.3) at each j. The original initial restriction has density $\mathbf{1}_{Y_R^*}/c_* \le 1/c_*$ with respect to ν; further exact labels only decrease the positive integral. Outside the null exact-tangency set, existence of a clearance witness implies $N_{2\varepsilon r^{d'_j}} \ge 1$. There are at most two flight indices of each nonnegative depth, so $\sum_j 2\varepsilon r^{d'_j} \le 4\varepsilon/(1-r)$. No independence or return-conditioned invariant measure is introduced. □

The estimate (A.5) is an integrated mass estimate. Multiplication of it by m^2 at fixed $\varepsilon(B)$ is not a proof about a target-roof density.

Appendix B. Clearance strips transverse to physical test curves

A measure bound alone gives no short-curve control. The positive separation between an unvisited disk and the incident disks supplies a transverse derivative which remains nonzero even as an incident angle approaches grazing. We specify the time orientation explicitly: forward near misses are tested on increasing-slope curves, and *past-flight* near misses on decreasing-slope curves.

Use physical arclength $s = R\alpha$ increasing counterclockwise and $p = \sin\varphi$. Fix a finite $U > 0$ and put

$$\begin{aligned} \kappa_- &= 1/R_+, \qquad \kappa_+ = 1/R_-, \\ D_0 &= \kappa_+ + U + (1 - R_+)^{-1}, \\ \ell_0 &= \pi/(2D_0), \qquad a_0 = \kappa_-(g_* - \delta_*), \\ C_{\mathrm{geo}} &= 2N_*\sqrt{1+U^2}/a_0. \end{aligned} \tag{B.1}$$

All lengths below are Euclidean arclength in (s, φ), not configuration-space arclength. A test is a connected graph over an s interval, of length at most ℓ_0. Let E_δ^+ be the forward physical near-miss event $\{N_\delta \ge 1\}$, and let E_δ^- be the analogous event on the preceding actual flight.

Theorem B.1 (A grazing-uniform, one-flight strip bound)**.** *For every* C^1 *test graph* W^+ *with* $0 \le d\varphi/ds \le U$*, and every* C^1 *test graph* W^- *with* $-U \le d\varphi/ds \le 0$*,*

$$|W^+ \cap E_\delta^+| \le C_{\mathrm{geo}}\delta, \qquad |W^- \cap E_\delta^-| \le C_{\mathrm{geo}}\delta. \tag{B.2}$$

The bound is uniform in R*, in the homogeneity strip of the test, and in its distance to the grazing boundary. Arbitrary Borel restrictions, including exact record restrictions, can be added to the sets on the left. Regular pieces are understood with their physical multiplicity one.*

Proof. On W^+ the direction angle is $\vartheta = s/R + \varphi(s)$. For a fixed candidate center $z \neq 0$ set

$$t_z = (z-q)\cdot v_\vartheta, \qquad b_z = (z-q)\cdot v_\vartheta^\perp.$$

Differentiation in the increasing s coordinate gives the signed identity

$$b_z' = -\cos\varphi - t_z\vartheta', \qquad \kappa_- \le \vartheta' \le \kappa_+ + U. \tag{B.3}$$

At a physical near miss, Lemma A.1 yields $t_z \ge g_* - \delta_*$. Hence $-b_z' \ge a_0$ there. It is the $t_z\vartheta'$ term, not a lower bound on $\cos\varphi$, which prevents degeneration.

We must not multiply this estimate by an uncontrolled number of components. Choose a continuous argument $\beta(s) = \arg(z - q(s))$ on the graph. Since $|z-q| \ge 1 - R_+$, one has $|\beta'| \le (1-R_+)^{-1}$. Moreover

$$(\beta - \vartheta)' \le (1-R_+)^{-1} - \kappa_- < 0,$$

because $R_+ < 1/2$. Its total variation is at most $D_0|W^+| \le \pi/2$. Thus the set $t_z > 0$, equivalently $\cos(\beta - \vartheta) > 0$, is at most one interval: two distinct positive-cosine intervals are separated by a negative interval of angular length π. On the interval $t_z > 0$, formula (B.3) says that b_z strictly decreases. Each of the two impact bands $(R, R+\delta)$ and $(-R-\delta, -R)$ therefore has multiplicity at most one in this coordinate. Restricting to the portions which are actual near misses can break them into pieces, but cannot increase that multiplicity. By the one-dimensional change of variables on this monotone map,

$$\int_{\{s:\ z \text{ is a physical near miss}\}} ds \le 2\delta/a_0.$$

Since $d\ell \le \sqrt{1+U^2}\, ds$, summing over the finite candidate set from Lemma A.1 proves the first bound. This sum counts only centers in one physical flight. It does not count iterated collision words or homogeneous continuity components.

For the second bound, the collision time-reversal involution is $\mathcal{I}(s,\varphi) = (s, -\varphi)$. The outgoing velocity at $\mathcal{I}x$ reverses the incoming velocity at x. Consequently its next straight flight is the reversal of the previous actual flight at x. This proves

$$E_\delta^- = \mathcal{I}E_\delta^+ = T_R E_\delta^+ \tag{B.4}$$

on the regular collision space. The involution is an isometry in (s,φ) and sends W^- to an increasing-slope graph. Apply the first bound. Borel restrictions decrease lengths. Taking monotone limits of truncated regular test pieces gives the same bound at grazing limits; an exactly tangent parameter contributes zero transverse length by the strict derivative above. This last statement does not delete any paired-source boundary trace in the older coarea construction. □

Corollary B.2 (Bounded multipliers on stable tests)**.** *If $|h| \le H$ pointwise on regular collision states, then for every W^- as above and every bounded Borel test ψ,*

$$\left| \int_{W^-} \mathbf{1}_{E_\delta^-} h\psi \, d\ell \right| \le H\|\psi\|_\infty \min\{|W^-|, C_{\mathrm{geo}}\delta\}.$$

This is a bound for bounded functions with specified representatives. It is not an assertion that multiplication by the indicator is bounded on an entire space of anisotropic distributions.

Proof. Bound the integrand by $H\|\psi\|_\infty \mathbf{1}_{E_\delta^-}$ and use Theorem B.1 together with the trivial length bound. Representatives on singular sets are fixed to zero. □

Appendix C. Clearance gains after arbitrarily many actual collisions

We next give a consequence which is uniform in the full collision count, rather than only a one-flight estimate. Its additional input is the retained stable-curve growth bound, stated explicitly here. It is not the stronger target-roof local limit estimate.

Input statement C.1 (Retained homogeneous curve growth)**.** Use a family $\mathcal{W}_R^s$ of short homogeneous admissible stable graphs in the above physical coordinates, with slopes in $[-U, 0]$, length at most ℓ_0, and the inherited common curvature bound. For every $W \in \mathcal{W}_R^s$ and $l \geq 0$, its regular inverse image has a countable partition $\mathcal{G}_l(W)$ into such graphs, up to arclength-null sets, with one-copy inverse branches and

$$\sum_{V \in \mathcal{G}_l(W)} \sup_V J_V T_R^l \leq C_g. \tag{C.1}$$

Here $J_V T_R^l$ is the Euclidean arclength derivative along V; C_g is independent of R, l, W. At $l = 0$ the partition is $\{W\}$.

This is the curve-growth interface of historical module 241, not its whole-cell pressure conclusion. Length thresholds can be decreased: subdividing a pre-existing short component into lengths at most ℓ_0 multiplies its Jacobian upper sum by at most a fixed number, independent of l. The precise external interface is discussed in Appendix G. We retain this input separately from the unconditional one-flight geometry of Theorems A.2 and B.1.

For fixed representatives of bounded Borel functions put

$$\mathcal{S}_a(f) = \sup_{W \in \mathcal{W}_R^s} |W|^{-a} \int_W |f| \, d\ell, \qquad 0 \leq a < 1.$$

This absolute stable-test seminorm is specified here to avoid identifying it with a full anisotropic distribution norm. With respect to the invariant measure ν, the ordinary density transfer is $\mathcal{L}_R h = h \circ T_R^{-1}$. For the original record observable $\mathbf{g}_R = (\kappa_1, \kappa_2, \tau, \mathbf{1}_{Y_R^*})$ and real $\xi = (\xi_1, \xi_2, \xi_t, \xi_n)$, define

$$\mathcal{L}_{R,\xi} h(x) = e^{i\xi \cdot \mathbf{g}_R(T_R^{-1}x)} h(T_R^{-1}x).$$

The three discrete frequencies may be taken modulo 2π; the physical-roof frequency is real and unbounded. This definition does not assert boundedness of this operator on an unstated Banach space. On bounded functions its modulus is simply $|h| \circ T_R^{-1}$.

Theorem C.2 (A uniform future-propagated clearance gain)**.** *Under Input C.1, let $|h| \leq H$ pointwise on the regular collision space, with $h = 0$ off E_δ^-. Then for every $l \geq 0$, real ξ, and $0 \leq a < 1$,*

$$\int_W |\mathcal{L}_{R,\xi}^l h| \, d\ell \leq H \min\{|W|, C_g C_{\mathrm{geo}} \delta\}, \tag{C.2}$$

$$\mathcal{S}_a(\mathcal{L}_{R,\xi}^l h) \leq H (C_g C_{\mathrm{geo}} \delta)^{1-a}. \tag{C.3}$$

The constants do not depend on the entire future collision word or on any of the four real frequencies. The bounds also hold for any bounded multiplier of modulus at most one applied at arbitrary intermediate collisions, with the same support at the marked time.

Proof. On regular branches positivity and $|e^{i\xi \cdot \mathbf{g}_R}| = 1$ give $|\mathcal{L}_{R,\xi}^l h(x)| \leq H \mathbf{1}_{E_\delta^-}(T_R^{-l}x)$. Change variables separately on the inverse branches of W. The full nonnegative sum, completed by monotone convergence, satisfies

$$\int_W |\mathcal{L}_{R,\xi}^l h| \, d\ell \leq H \sum_{V \in \mathcal{G}_l(W)} (\sup_V J_V T_R^l) \, |V \cap E_\delta^-| \leq H C_{\mathrm{geo}} \delta C_g.$$

The pointwise bound by H supplies the other term in the minimum. For $x > 0$ and $D > 0$, splitting into $x \le D$ and $x \ge D$ gives $x^{-a}\min(x, D) \le D^{1-a}$. Apply this with $x = |W|$ and $D = C_g C_{\mathrm{geo}}\delta$ to obtain (C.3). The modulus of a product of additional multipliers is at most one, so precisely the same positive upper bound applies. No stochastic independence, wordwise supremum in place of a physical integral, or Fourier inversion has been used. □

Corollary C.3 (The complete depth schedule in stable tests)**.** *Let $f_0 = \mathbf{1}_{Y_R^*}/c_*$. For $0 \le j < m$ set*

$$h_j = \mathbf{1}_{E^-_{\delta_j}} \mathcal{L}_{R,\xi}^{j+1} f_0, \qquad f_{m,j} = \mathcal{L}_{R,\xi}^{m-j-1} h_j.$$

Then

$$\mathcal{S}_a(f_{m,j}) \le c_*^{-1}(C_g C_{\mathrm{geo}}\delta_j)^{1-a}. \tag{C.4}$$

If $2\varepsilon \le \delta_*$ and $\delta_j = 2\varepsilon r^{d'_j}$, the terminal collision density restricted to the union of all original clearance witnesses has absolute stable-test seminorm at most

$$\frac{2(C_g C_{\mathrm{geo}})^{1-a}(2\varepsilon)^{1-a}}{c_*(1 - r^{1-a})}, \qquad 0 \le a < 1. \tag{C.5}$$

This holds uniformly over m, R and real record frequencies. Further original exact labels and terminal-section restrictions may be retained on the left by positive domination.

Proof. The initial density and all of its twisted iterates have modulus at most $1/c_*$. By (B.4), the past-flight event at collision $j+1$ is exactly the forward clearance mark on flight j. Thus the exponents $j+1$ and $m-j-1$ insert the mark at the actual collision time, including $j = 0$ and $j = m-1$ with zero remaining future steps. Theorem C.2 proves (C.4). The union indicator is bounded by the sum of its physical one-flight indicators. Subadditivity of $\mathcal{S}_a$ and at most two indices at each depth give

$$\sum_{j=0}^{m-1} \delta_j^{1-a} \le \frac{2(2\varepsilon)^{1-a}}{1 - r^{1-a}}.$$

This proves (C.5). Arbitrary exact-record selectors are bounded by one on the same trajectories, so they cannot increase the positive majorant. They are not replaced by independent weights, and no probability measure is renormalized. □

The estimate is genuinely uniform in the full number of physical collisions and in the full scheduled set of clearance marks. Its norm is nevertheless a stable-test norm on terminal collision space, not the essential height of a roof density. In particular, the absence of frequency growth in this absolute estimate is not decay as $|\xi_t| \to \infty$ and does not justify integration over that unbounded frequency. Such decay, or another rigorously justified pointwise bridge, is still required for the original raw target.

Appendix D. Exact-record disintegration at an unvisited mark

The flux coordinates also give a direct representation of the remaining roof problem which does not discard the exact return, displacement or collision information. We state it as an equality of measures first, because a conditional angular roof kernel need not have a density without proof.

For $0 \le j < m$ define the initial state

$$X_{R,j}(\vartheta, \sigma, u) = T_R^{-j} X_R(\vartheta, \sigma, u)$$

on its regular set, with a fixed null-set convention otherwise. The complete actual record from this state is used. Displacements are computed in the original lattice basis, relative to its initial

lift; translating the marked disk to zero changes none of these relative displacements. For almost every u, define the positive Borel measure on the original roof line

$$\mathcal{K}_{m,R,n,k,j,u}(A) = \sum_{\sigma=\pm 1} \int_0^{2\pi} \mathbf{1}_{\Pi_{n,k,m,R}}(X_{R,j}) \mathbf{1}_{\{L_{m,R}(X_{R,j})\in A\}}\, d\vartheta. \tag{D.1}$$

It has mass at most 4π. No density or angular derivative bound is included in this definition.

Theorem D.1 (Disintegration of a physically marked exact record)**.** *For every Borel set $A \subset \mathbb{R}$,*

$$\int \mathbf{1}_{\Pi} N_\delta(T_R^j x) \mathbf{1}_{\{L_{m,R}\in A\}}\, d\nu_R^* = \frac{1}{4\pi R c_*} \int_0^\delta \mathcal{K}_{m,R,n,k,j,u}(A)\, du. \tag{D.2}$$

The identity also holds against every nonnegative Borel roof test. All singular boundary sets in these integrals use the original collision-area null-set convention, not deletion of paired boundary coefficients.

Proof. Replace $d\nu_R^*$ by $c_*^{-1} d\nu$ inside the integral, since $\Pi \subset Y_R^*$. Change the initial collision coordinate to $y = T_R^j x$ using invariance of ν. Apply Theorem A.2 with

$$a(y,z) = \mathbf{1}_\Pi(T_R^{-j} y) \mathbf{1}_{\{L_{m,R}(T_R^{-j} y)\in A\}}.$$

This is a bounded nonnegative function on the original physical states, independent of which near-miss mark is selected once y is fixed. The resulting integral is exactly (D.2). Countable regular partitions and Tonelli's theorem justify the integration over u; their exceptional parameters have two-dimensional measure zero. For arbitrary nonnegative roof tests use monotone simple-function approximation. In particular the factor is $1/(4\pi R c_*)$, with no second section normalization and no new conditioning on the near miss. □

D.1. **A sufficient pointwise bridge, with its status explicit.** Fix M. The following measure domination is one concrete, stronger sufficient condition for paying all original clearance witnesses: there exists C_M such that, for all sufficiently large m, all R, n, k, j and almost every $0 < u < \delta_*$,

$$\mathcal{K}_{m,R,n,k,j,u}\big|_{\mathcal{C}_{m,R,n,k}(M)} \le C_M m^{-2}\, dt. \tag{D.3}$$

An almost-everywhere set may depend on the fixed R, m, n, k, j; C_M must be common. This is an *unproved sufficient condition*, not a newly assumed premise of the organizing raw local limit assertion. It asserts absence of singular conditional roof mass as well as an upper density bound. Neither Theorem A.2 nor Theorem C.2 establishes it.

Proposition D.2 (The exact normalization a pointwise bridge must pay)**.** *If* (D.3) *holds, the original roof measure restricted to any clearance witness, with $2\varepsilon \le \delta_*$, is dominated on the central target by a density d_{clr} with*

$$\sup_{R,n,k} m^2 \|d_{\mathrm{clr}}\|_{L^\infty(\mathcal{C}_{m,R,n,k}(M))} \le \frac{C_M \varepsilon}{\pi R_- c_* (1-r)}. \tag{D.4}$$

The same upper bound holds for any positive source already dominated by that original collision measure, restricted to these witnesses.

Proof. The witness-union indicator is at most $\sum_j N_{2\varepsilon r^{d_j'}}(T_R^j x)$ off the fixed null set. Apply (D.2) to every term, then the measure domination (D.3). The resulting density bound is $(4\pi R c_*)^{-1} C_M m^{-2} \sum_j 2\varepsilon r^{d_j'}$. The sum is at most $4\varepsilon/(1-r)$, proving the stated constant. Positive domination preserves the bound for the indicated source. At the original outer schedule this would tend to zero after the collision limsup and then $B \to \infty$; no window has been changed. □

On a regular physical branch, the new parametrization has the exact chain rule

$$DX_{R,j} = DT_R^{-j}(X_R)\, DX_R, \qquad D(L_{m,R} \circ X_{R,j}) = DL_{m,R}(X_{R,j})\, DX_{R,j}.$$

The one-flight lower bound for $|b_z'|$ does not lower-bound this long-time roof derivative. Inverse-iterate growth, cancellations and critical angular roofs must be controlled separately. A bound after integration in u can allow angular folds which a uniform bound on each $\mathcal{K}_{j,u}$ would exclude. No supremum is passed through that disintegration here.

A proof of the stronger condition (D.3) is not necessary if a suitably summed kernel bound or comparison-capacity argument can be proved instead. The proposition only makes explicit which normalization and which measure must be controlled. It is not counted as an unconditional closure or a second payment of a source already paid in revision 112.

D.2. **Compatibility with the complete original source.** All identities through (14.3) and the full pool $P_{r_B}^{\mathrm{ng}}$ are unchanged. We have not subtracted the clearance source from $\mathsf{U}_B^{\mathrm{low}}$ using only a weak norm. Consequently the exact remaining target is still (14.8), with the original order of limits. The new results treat the mechanism which is invisible when $J_\varepsilon = \varnothing$: genuine unvisited-disk near misses. They do not pay low-severity shallow incidence without clearance, prove abundance of the full nongrazing comparison pool, or settle positive critical-entrance and section-cut pairing traces.

For precision concerning a minor referee point, the inequality $\alpha f(N) < u_m$ already proved in Corollary 14.2 implies $N < 2\sqrt{u_m/\alpha}$ on the retained source. The independent quadratic count theorem is retained as a finite-count theorem, not presented as another missing payment for that already restricted source. The term "low" in all inherited source symbols means low *incidence* load only. In the monotone exhaustion of Proposition 9.4, every marked weight is explicitly assigned zero on the omitted collision-area null set; its extended integral is taken only over the stated regular charts. This convention never sets a positive one-sided pairing trace to zero.

Appendix E. Scope of the retained comparison statements

For clarity, we record the precise scope of the earlier allocation statements. None of their algebraic maximality assertions is used as a substitute for the physical estimate in Theorem 6.1.

Proposition E.1 (Fixed-window scalar maximality and old-family dominance)**.** *Fix a Borel comparison subspace, $0 < r < \Delta$, and $K > 0$. Consider nonnegative measurable weights w_+, w_- with $w_+ + w_- \le 1$, supported on the respective one-sided window conditions in* (5.2)*. Among scalar accepted heights $Z(t)$ satisfying*

$$0 \le Z(t) \le U(t), \qquad Z(t) \le K \int (w_+ + w_-) p_\vee \, d\theta,$$

the greatest is $V_{r,K}(t) = \min\{U(t), KP_r(t)\}$. The allocation is global at the roof and does not couple individual source sheets to comparison sheets.

For a compatible old fixed family with window width r_{old}, physical overlap N and block cost κ, the explicit conditions

$$0 < r \le r_{\mathrm{old}}, \qquad \ell_{\mathrm{comparison}} < \Delta, \qquad K = N\kappa$$

imply $V^{\mathrm{old}}(t) \le V_{r,N\kappa}(t)$ on the same original source and comparison space.

Proof. The support and one-copy conditions give $w_+ + w_- \le Q_r$. Hence every Z is bounded by the stated minimum. On $P_r > 0$ choose $w_\pm = V_{r,K} q_\pm / (KP_r)$; use zero weights otherwise. This is a measurable attaining certificate. The separate source multiplier is $\beta_{r,K}$ defined in Section 6. For the old family, retain only active source blocks at t. Their comparison supports lie in the r-pool by the displayed width condition. Summing the blockwise budget and using physical

overlap at most N gives $V^{\mathrm{old}} \le N\kappa P_r$; disjoint original source blocks also give $V^{\mathrm{old}} \le U$. The minimum proves the assertion, with countable families obtained by monotone convergence on their common roof versions. □

Proposition E.2 (Strictness relative to roof-independent whole-interval blocks)**.** *In the interval-source model, the roofwise certificate class is strictly larger than the old class of roof-independent angular source blocks that requires a comparison interval to contain the translate of the* entire *source interval by* $[0, r]$ *or* $[-r, 0]$*. This statement compares certificate classes and does not assert a Lorentz realization.*

Proof. Let $s = 1/1000$ and $r = 1/10$. On two disjoint angular sets of measure one give both endpoint traces value one. Let their physical intervals be $(0, s)$ and $(3s/8, r + 7s/8)$, and select only the first as source. Take $\Delta > r + s/2$ and $K = 1$. For $3s/8 < t < 7s/8$ the second interval contains $[t, t + r]$, so $P_r(t) \ge U(t) = 1$. Neither physical interval has length $r + s$, which is necessary for the old translated-whole-source inclusion. Angular subdivision does not alter these intervals. Thus the old class cannot allocate a nonzero whole-source block while the roofwise class accepts the source on the indicated roof interval. □

E.1. **Limit order and the remaining estimate.** The constructions in this article are evaluated first at finite m, R, n, k with B, r_B, h_B, K_B already fixed. Positive physical sums are completed before taking the essential roof supremum, the radius and exact-label suprema, and the collision limsup. Only then is B sent to infinity. The assertion that every positive-length segment contains a sufficiently small window at a fixed count supplies no uniform modulus for this order.

The remaining estimate is precisely

$$\lim_{B\to\infty} \limsup_{m\to\infty} \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} \left(\mathsf{U}^{\mathrm{micro}}_{B,m}(t) - K_B P^{\mathrm{ng}}_{r_B,m}(t)\right)_+ = 0.$$

No inference to this estimate is made from bounded wordwise optical weights, local integrability, a finite diagnostic, or a positive finite-count incidence. Theorem 6.1 proves the entire visited-grazing payment in this order; all remaining positive classes in the displayed expression are still part of the original pointwise problem.

Appendix F. Optical cofactors, multiple marks and simultaneous endpoints

We supply explicit matrix details for the optical identity and isolate the one-sided endpoint statement used in the trace theorem. These calculations take place on the original finite itinerary. They do not assert a bound for a sum over itineraries.

Lemma F.1 (The signed corner cofactor)**.** *For* $m \ge 1$*, with the empty determinant convention when* $m = 1$*,*

$$G_{0m} = (-1)^{m-1} \frac{R^2 c_0 c_m}{(\prod_{i=0}^{m-1} \tau_i) \det J}. \tag{F.1}$$

In particular the absolute value, not the signed mixed entry, gives $\omega = |G_{0m}|/R$*.*

Proof. Suppose first that $m > 1$ and put $n = m - 1$. The minor obtained by deleting the last row and first column of the tridiagonal matrix J is triangular, with diagonal entries $\tau_1^{-1}, \dots, \tau_{n-1}^{-1}$. Cramer's rule therefore gives

$$(J^{-1})_{1n} = (-1)^{n+1} \frac{\prod_{i=1}^{n-1} \tau_i^{-1}}{\det J}.$$

For $n = 1$ its numerator is the empty product and the formula says $(J^{-1})_{11} = 1/J_{11}$. The only endpoint–interior couplings are $H_{01} = R^2 c_0 c_1/\tau_0$ and $H_{nm} = R^2 c_n c_m/\tau_{m-1}$, whereas $H_{0m} = 0$.

Since $H_{II}^{-1} = R^{-2}C_I^{-1}J^{-1}C_I^{-1}$, the Schur complement gives

$$G_{0m} = -\frac{R^2 c_0 c_m}{\tau_0 \tau_{m-1}}(J^{-1})_{1n},$$

which proves (F.1). If $m = 1$, the mixed entry follows directly from the single square in (3.2) and has positive sign.

For clarity, the equality between the determinant denominator and $\mathfrak{b}_m$ can also be checked without a sign convention for optical coordinates. Both quantities have initial values $B_0 = 0$, $B_1 = \tau_0$ and satisfy

$$B_{l+1} = \left(1 + \tau_l\kappa_l + \frac{\tau_l}{\tau_{l-1}}\right) B_l - \frac{\tau_l}{\tau_{l-1}} B_{l-1}.$$

For the determinant this is the tridiagonal recurrence after multiplication by the flight lengths. For the optical product it follows by eliminating the lower entry of its second column. This verifies the denominator identity used in Lemma 3.3. □

Put $T_{a,b} = \sum_{i=a}^{b-1} \tau_i$ for $0 \le a < b \le m$. For $S = \{j_1 < \cdots < j_s\} \subset \{1, \dots, m-1\}$ set $j_0 = 0$ and $j_{s+1} = m$.

Proposition F.2 (Positive expansion and simultaneous optical marks)**.** *The exact denominator has the positive expansion*

$$\mathfrak{b}_m = \sum_{S \subset \{1,\dots,m-1\}} \left(\prod_{j \in S} \kappa_j\right) \prod_{\nu=0}^{|S|} T_{j_\nu, j_{\nu+1}}. \tag{F.2}$$

For every fixed set S of s distinct interior marks,

$$\frac{\omega}{\prod_{j \in S} c_j} \le \frac{R_+^{s+1}}{2^s \tau_-^{s+1}} \varrho^{m-s-1}. \tag{F.3}$$

For $S = \varnothing$ this is the first optical bound in (3.8)*; for $|S| = 1$ it is the second.*

Proof. Let $E = \left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)$. Expand every factor $\mathsf{C}(\kappa) = I + \kappa E$ in the product (3.6). Between two selected marks the shears combine according to $\mathsf{S}(a)\mathsf{S}(b) = \mathsf{S}(a+b)$, and $E\mathsf{S}(T)E = TE$. The top-right entry of the term with marked set S is exactly the corresponding summand in (F.2). The empty term is $T_{0,m}$. Every coefficient is positive.

To obtain the stronger estimate retaining the unmarked reflections, expand only the factors at the prescribed marks. The coefficient of $\prod_{j \in S} \kappa_j$, with all other factors retained, is the product of the top-right entries of the $s+1$ intervening optical blocks. Indeed $E = e_2 e_1^{\mathsf{T}}$, so repeated multiplication factors that coefficient into these entries. The block with $j_{\nu+1} - j_\nu$ flights has top-right entry at least $\tau_- \varrho^{-(j_{\nu+1} - j_\nu - 1)}$, by the same positive-column argument as in (3.9). Keeping only this nonnegative term gives

$$\mathfrak{b}_m \ge \left(\prod_{j \in S} \frac{2}{R c_j}\right) \tau_-^{s+1} \varrho^{-(m-s-1)}.$$

Substitute this in $\omega = R c_0 c_m / \mathfrak{b}_m$, and use $c_0 c_m \le 1$ and $R \le R_+$, to prove (F.3). □

This estimate controls simultaneous vanishing incidences on a fixed finite trajectory. Its constant depends on the number of marks, not on their locations. No word-count factor has been suppressed, and no pressure conclusion is drawn by summing its right side.

Lemma F.3 (One-sided limits at simultaneous physical mechanisms)**.** *Fix a positive-length pairing segment and approach one of its ends from the regular interior. The collision points, velocities, flight lengths, visited incidences and third-disk clearances have limits along the entire approach, even if several of the following occur simultaneously: visited grazing, unvisited tangency, a section boundary and a critical incoming endpoint. The limiting reduced Hessian and coarea factor are those in* (4.3) *and* (4.4)*. No continuation of the exact event through its boundary is asserted.*

Proof. Lemma 4.1 gives limits for q and z along the entire segment, not only a subsequence. Hence the initial collision point and $p_0 = -z\cos\theta/R$ converge. The section cap gives the outgoing normal component $c_0 > 0$ at the limit, so the initial velocity converges as well.

Suppose inductively that the departure point x and unit velocity v of a designated flight converge. For its designated next disk with centre O, the entry time is

$$\tau = -(x-O)\cdot v - \sqrt{((x-O)\cdot v)^2 - |x-O|^2 + R^2}.$$

This is the entry root on the actual regular itinerary. Its radicand is nonnegative along the approach and has a nonnegative limit. The root is continuous also when the radicand tends to zero; $\tau \geq \tau_- > 0$ prevents a zero-length flight. The arrival point therefore converges. Reflection by $v \mapsto v - 2(v\cdot n)n$ is continuous even at tangency, so the next velocity converges. Induction applies a finite number of times. An earlier disk may become tangent only at the limiting flight; this does not alter the designated entry-root limit taken from the original event.

At fixed itinerary and radius, the bounded flight lengths restrict all possibly nearby third-disk centres to a finite set. Distance from a compact line segment to each such disk is continuous in the endpoints. Their finite minimum, and hence the original clearance, has a limit. A section decision may change at the limiting state, but its coordinate limit is still obtained from the same side. None of these continuity arguments requires the boundary mechanisms to occur separately. Formula (4.3), with Z^{-1} uniformly bounded, then gives the Hessian limit even with several vanishing incidences. The fixed-direction coarea formula gives the asserted limit for χ. □

Corollary F.4 (An explicit uniform trace modulus)**.** *A permissible constant in* (4.6) *is*

$$C_{\mathrm{H}} = \frac{500 C_{\mathrm{reg}}\sqrt{2\Lambda_\circ}}{4\pi c_* \lambda_\circ^2}. \tag{F.4}$$

It is common to all original sheets, collision counts, radii and exact labels. Its inherited dependence is precisely the regular variation constant C_{reg}; the remaining constants are the fixed geometry and section constants displayed in this article.

Proof. Corollary 3.4 and $\chi \geq \lambda_\circ$ give $AI_m \leq 500\varrho^{(m-2)_+}/(4\pi c_*\lambda_\circ)$. Integrate the regular derivative bound between u and v, and apply Lemma 4.1 to that subsegment:

$$\int_u^v z^{-1}\,ds \leq \lambda_\circ^{-1}\sqrt{2\Lambda_\circ(v-u)}.$$

Multiply the constants and pass to the already proved endpoints. □

Any nonempty collection of visited endpoint grazings belongs to Γ, and one vanishing marked incidence is already sufficient for its zero source statement. Proposition F.2 additionally controls their simultaneous product. This does not alter the distinct treatment of an unvisited third-disk tangency.

Appendix G. Dependency, reference and normalization audit

The direct clearance proofs use the original positive disk separation, the uniform finite-horizon hypothesis, circular curvature, elastic reflection and elementary change of variables. The arbitrary-time stable-test estimate additionally uses Input C.1. Neither uses uniform pressure, a twisted spectral gap, or the unproved localized-shortage estimate to prove its own premise.

G.1. **The precise imported curve estimate.** In the inspected 10 October 2013 author version of Demers–Zhang [6], Lemma 3.2(b), printed page 14, bounds the sum of suprema of the forward arclength Jacobians on homogeneous inverse-image pieces, uniformly for W and for maps in the stipulated class. The geometric-family statement is Theorem 2.7 in that version; the varying-arclength qualification is Remark 2.9(b). The inherited module 241's numbering must therefore be matched to a specified version rather than silently treated as verified. Its stronger finite-testing and pressure argument is not needed for Theorem C.2.

For this paper the coordinates, Euclidean arclength Jacobian, short-curve curvature bound, homogeneous partition, fixed subdivision length, and map-family constants must all agree with Input C.1. Changing boundary perimeter with R requires the coordinate transport and its uniform equivalence; changing the subdivision threshold changes C_g by a bounded factor. These interfaces remain explicit retained hypotheses here, rather than a claim of an independent specialist reconstruction.

In the inspected 19 September 2022 author version of Baladi–Demers [7], Lemma 2.2, printed pages 7–8, is a distortion statement across stable curves in the same homogeneous backward cell. Its closure convention excludes true singularity curves. It is not a four-record, roof-conditioned m^{-2} estimate. The present proofs make no such inference.

G.2. **Checks which enter the proofs.** The radius factor in (A.4) is R, not R^2 or $1/R$. The two impact signs and the full direction circle give the factor $4\pi\delta$ before collision normalization. Exactly one section factor c_*^{-1} occurs in (D.2). The foot has physical time coordinate t_z, so the sign in (B.3) is a minus for both the boundary motion and the direction motion. Increasing-slope curves, not forward stable curves, are used for the forward event.

The uniform lower bounds used above are strictly positive on the whole closed radius interval:

$$g_* - \delta_* = 9/200, \qquad \kappa_- - (1 - R_+)^{-1} = 600/2491, \qquad a_0 = 9/94.$$

The occurrence of N_* in the curve estimate, but not in the exact flux identity, is deliberate: the former is a one-flight union bound on a test curve, whereas the latter is a global one-copy change of section. Empty future blocks are included by $l = 0$. The denominator $1 - r^{1-a}$ is finite only for $a < 1$; no estimate at the endpoint $a = 1$ is claimed.

The source identities in the retained proof bodies are unchanged. All positive sums are completed before essential suprema and the collision limsup. Taking $B \to \infty$ in a stable-test estimate would not by itself permit taking that limit in the target-roof norm. Fixed-m simulations and symbolic identities check algebra and implementation only; they do not certify either the inherited uniform dynamical inputs or the remaining raw pointwise theorem.

G.3. **What is retained, and what has not been reconstructed.** The complete active revision-112 manuscript and its ten active core proof bodies are retained in this manuscript, and an unmodified copy of the downloaded active packet is supplied in `provenance/v112-active`. The original repository additionally contains the recursive revision-111 archive, with source tree `790e627efe3a89a4549123d7e9acacf24089c287`. That recursive archive was not exported by the completed CI artifact available for this revision. Its identity is preserved; it has not been represented as a locally reconstructed, independently audited set of 267 historical proofs. The included source manifest and preservation audit distinguish byte-verified local preservation

from remote tree pinning. In particular the historical spectral, arithmetic, induced-return and four-coordinate inverse-problem inputs are not replaced by the new clearance lemmas.

Appendix H. Critical-roof and fold audit: normalization and ordered closure

The new identities have different kinds of uniformity. The clearance-to-endpoint determinant is an identity for each finite physical itinerary and every marked flight. The fold and finite-type bounds are uniform over Borel selectors on a specified chart, with specified derivative constants. The critical coarea error is uniform over selectors but depends on the chart's third-derivative and weight-modulus constants. The critical coefficient sum is over all regular physical words at fixed collision count, relative to the retained uniform pressure input; it is unscaled. None of these qualifiers can be replaced by another.

H.1. **Two distinct uses of reversibility.** At a normal endpoint, the actual reflected ray reverses its incoming velocity. This proves the reversing-periodic-orbit assertion without adding a reflection at an unvisited mark. In collision coordinates, time reversal also gives the derivative identity $\mathcal{M} = JB^{-1}JB$. The first assertion concerns actual points and their section tests; the second concerns the tangent map on the regular domain. At an unvisited tangency, only the one-sided reduced action coefficient is used. A smooth extension of that action does not define the physical map through the forbidden side.

Even at a regular reversing orbit the original section need not be reversal invariant. The occupation identity is $A_{2m} = A_m + A_m^{\mathcal{I}}$ when both original endpoints are in the section, not automatically $2n$. The coefficient $\mathfrak{J}$ describes nearby collision area pushed to the roof. It is not a conditional law on a measure-zero periodic orbit.

H.2. **The exact small-scale error and its range.** The radius r_0 in Theorem 16.3 is a local endpoint-chart radius. The height increment is $s = t - t_x$, and its allowed range is $0 < s < r_0^2/3$. Neither r_0 nor s replaces the original microscopic observation window. In particular the explicit error constant

$$2\pi\sqrt{3}\{2L_w + Kw(0)\}$$

has not been shown summable after multiplying the full physical source by m^2. Decreasing r_0 separately for each orbit does not establish that summability. A general Borel selector retains the exact fraction $a_E(s)$; only a verified finite tangent-sector structure justifies a limiting angular fraction.

The local fold exponent $1/2$ is produced by integrating a square-root conditional singularity across a transverse clearance direction. A higher finite type has its stated exponent $1/\ell$ and derivative budget. Full critical roofs have a different two-dimensional local coarea law; the fold theorem is not applied there when its clearance transversality fails. A countable disjoint source-chart covering preserves positivity and exact multiplicity, but does not itself bound the sum of the chart constants.

H.3. **The unchanged closure statement.** All original positive coefficients and the complete nongrazing pool are still those of Section 14. The remaining statement is (14.8), with the full physical sum first, then the exact-label and radius suprema and the original roof essential supremum, then the collision limsup, and the outer limit last. A successful continuation must control clearance and critical charts, positive one-sided pairing traces, the low-incidence source, and the actual available comparison capacity in that topology. The new calculations neither subtract these unpaid classes nor introduce a new name for the shortage.

The accompanying programs check the signed generating derivative, stability determinant, normalization, fold constants, selector error and actual horizontal period-two physical rays. The latter are not asserted to lie in the large-collision central roof window. Exact rational

checks of their optical Schur complements and finite ray-derivative comparisons test independent implementations; they are not substitutes for the written continuous proofs or the unresolved uniform dynamical estimates.

References

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[7] V. Baladi and M. F. Demers, *Thermodynamic formalism for dispersing billiards*, Journal of Modern Dynamics **18** (2022), 415–493. The inspected author version is dated 19 September 2022.